

pdf-dc99cf88 — formula report

344 inline formulas, 46 display equations, 0 tables, 4 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
pdf-dc99cf88_EQ00021	010	■0.263	$\begin{aligned} \hat{\rho}_{\mathrm{dog}} &= \sum_{(a,b) \in A \times B} \pi(a,b, \text{dog}, \text{barks}) \text{barks} \otimes \text{barks}^* \\ &+ \sum_{(a,b) \in A \times B} \sqrt{\pi(a,b, \text{dog}, \text{barks}) \pi(a,b, \text{dog}, \text{runs})} \text{barks} \otimes \text{runs}^* + \\ &+ \sum_{(a,b) \in A \times B} \sqrt{\pi(a,b, \text{dog}, \text{runs}) \pi(a,b, \text{dog}, \text{barks})} \text{runs} \otimes \text{barks}^* + \\ &+ \sum_{(a,b) \in A \times B} \pi(a,b, \text{dog}, \text{runs}) \text{runs} \otimes \text{runs}^* \\ &= \frac{1}{5} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}. \end{aligned}$	$\hat{\rho}_{\mathrm{dog}} = \sum_{(a,b) \in A \times B} \pi(a,b, \text{dog}, \text{barks}) \text{barks} \otimes \text{barks}^* + \sum_{(a,b) \in A \times B} \sqrt{\pi(a,b, \text{dog}, \text{barks}) \pi(a,b, \text{dog}, \text{runs})} \text{barks} \otimes \text{runs}^* + \sum_{(a,b) \in A \times B} \sqrt{\pi(a,b, \text{dog}, \text{runs}) \pi(a,b, \text{dog}, \text{barks})} \text{runs} \otimes \text{barks}^* + \sum_{(a,b) \in A \times B} \pi(a,b, \text{dog}, \text{runs}) \text{runs} \otimes \text{runs}^* = \frac{1}{5} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}.$	$\hat{\rho}_{\mathrm{dog}} = \sum_{(a,b) \in A \times B} \pi(a,b, \text{dog}, \text{barks}) \text{barks} \otimes \text{barks}^* + \sum_{(a,b) \in A \times B} \sqrt{\pi(a,b, \text{dog}, \text{barks}) \pi(a,b, \text{dog}, \text{runs})} \text{barks} \otimes \text{runs}^* + \sum_{(a,b) \in A \times B} \sqrt{\pi(a,b, \text{dog}, \text{runs}) \pi(a,b, \text{dog}, \text{barks})} \text{runs} \otimes \text{barks}^* + \sum_{(a,b) \in A \times B} \pi(a,b, \text{dog}, \text{runs}) \text{runs} \otimes \text{runs}^* = \frac{1}{5} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}.$
pdf-dc99cf88_EQ00009	006	■0.272	$\psi = \frac{1}{2}(\text{big} \otimes \text{mountain} + \text{tall} \otimes \text{mountain} + \text{cold} \otimes \text{winter} + \text{chilly} \otimes \text{winter})$	$\psi = \frac{1}{2}(\text{big} \otimes \text{mountain} + \text{tall} \otimes \text{mountain} + \text{cold} \otimes \text{winter} + \text{chilly} \otimes \text{winter})$	$\psi = \frac{1}{2}(\text{big} \otimes \text{mountain} + \text{tall} \otimes \text{mountain} + \text{cold} \otimes \text{winter} + \text{chilly} \otimes \text{winter})$
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 — no value					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
pdf-dc99cf88_E00020	010	■ 0.342	$\begin{aligned} \rho_D &= \sum_{(a,b,c) \in A \times B \times C} \pi(a,b,c, \text{barks}) \text{barks} \otimes \text{barks}^* \\ &+ \sum_{(a,b,c) \in A \times B \times C} \sqrt{\pi(a,b,c, \text{barks}) \pi(a,b,c, \text{runs})} \text{barks} \otimes \text{runs}^* \\ &+ \sum_{(a,b,c) \in A \times B \times C} \sqrt{\pi(a,b,c, \text{runs}) \pi(a,b,c, \text{barks})} \text{runs} \otimes \text{barks}^* \\ &+ \sum_{(a,b,c) \in A \times B \times C} \pi(a,b,c, \text{runs}) \text{runs} \otimes \text{runs}^* \\ &= \frac{1}{5} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \end{aligned}$	$\rho_D = \sum_{(a,b,c) \in A \times B \times C} \pi(a,b,c, \text{barks}) \text{barks} \otimes \text{barks}^* + \sum_{(a,b,c) \in A \times B \times C} \sqrt{\pi(a,b,c, \text{barks}) \pi(a,b,c, \text{runs})} \text{barks} \otimes \text{runs}^* + \sum_{(a,b,c) \in A \times B \times C} \sqrt{\pi(a,b,c, \text{runs}) \pi(a,b,c, \text{barks})} \text{runs} \otimes \text{barks}^* + \sum_{(a,b,c) \in A \times B \times C} \pi(a,b,c, \text{runs}) \text{runs} \otimes \text{runs}^* = \frac{1}{5} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$	$\rho_D = \sum_{(a,b,c) \in A \times B \times C} \pi(a,b,c, \text{barks}) \text{barks} \otimes \text{barks}^* + \sum_{(a,b,c) \in A \times B \times C} \sqrt{\pi(a,b,c, \text{barks}) \pi(a,b,c, \text{runs})} \text{barks} \otimes \text{runs}^* + \sum_{(a,b,c) \in A \times B \times C} \sqrt{\pi(a,b,c, \text{runs}) \pi(a,b,c, \text{barks})} \text{runs} \otimes \text{barks}^* + \sum_{(a,b,c) \in A \times B \times C} \pi(a,b,c, \text{runs}) \text{runs} \otimes \text{runs}^* = \frac{1}{5} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$
pdf-dc99cf88_E00045	016	■ 0.432	$\mathcal{L}(s, s') = \begin{cases} \pi(s' s) & \text{if } s \leq s' \\ 0 & \text{otherwise} \end{cases}$	$\mathcal{L}(s, s') = \begin{cases} \pi(s' s) & \text{if } s \leq s' \\ 0 & \text{otherwise} \end{cases}$	$\mathcal{L}(s, s') = \begin{cases} \pi(s' s) & \text{if } s \leq s' \\ 0 & \text{otherwise} \end{cases}$
pdf-dc99cf88_E00008	006	■ 0.526	$T = \{(\text{big}, \text{mountain}), (\text{tall}, \text{mountain}), (\text{cold}, \text{winter}), (\text{chilly}, \text{winter})\}.$	$T = \{(\text{big}, \text{mountain}), (\text{tall}, \text{mountain}), (\text{cold}, \text{winter}), (\text{chilly}, \text{winter})\}.$	$T = \{(\text{big}, \text{mountain}), (\text{tall}, \text{mountain}), (\text{cold}, \text{winter}), (\text{chilly}, \text{winter})\}.$
pdf-dc99cf88_E00042	015	■ 0.568	$\hat{\rho}_{\text{dog}} = \sum_{i_2} \hat{\rho}_{x_{i_2} \text{dog}} = \sum_{i_3, i_2} \hat{\rho}_{x_{i_3} x_{i_2} \text{dog}} = \cdots = \sum_{i_{N-1}, \dots, i_3, i_2} \hat{\rho}_{x_{i_{N-1}} \cdots x_{i_3} x_{i_2} \text{dog}}.$	$\hat{\rho}_{\text{dog}} = \sum_{i_2} \hat{\rho}_{x_{i_2} \text{dog}} = \sum_{i_3, i_2} \hat{\rho}_{x_{i_3} x_{i_2} \text{dog}} = \cdots = \sum_{i_{N-1}, \dots, i_3, i_2} \hat{\rho}_{x_{i_{N-1}} \cdots x_{i_3} x_{i_2} \text{dog}}.$	$\hat{\rho}_{\text{dog}} = \sum_{i_2} \hat{\rho}_{x_{i_2} \text{dog}} = \sum_{i_3, i_2} \hat{\rho}_{x_{i_3} x_{i_2} \text{dog}} = \cdots = \sum_{i_{N-1}, \dots, i_3, i_2} \hat{\rho}_{x_{i_{N-1}} \cdots x_{i_3} x_{i_2} \text{dog}}.$
pdf-dc99cf88_E00039	014	■ 0.613	$\begin{aligned} \text{I climbed} &\leq \text{I climbed the tall mountain} \\ \text{a hot summer} &\leq \text{iced tea on a hot summer day} \\ \text{dog} &\leq \text{black dog} \leq \text{small black dog} \end{aligned}$	$\begin{aligned} \text{I climbed} &\leq \text{I climbed the tall mountain} \\ \text{a hot summer} &\leq \text{iced tea on a hot summer day} \\ \text{dog} &\leq \text{black dog} \leq \text{small black dog} \end{aligned}$	$\begin{aligned} \text{I climbed} &\leq \text{I climbed the tall mountain} \\ \text{a hot summer} &\leq \text{iced tea on a hot summer day} \\ \text{dog} &\leq \text{black dog} \leq \text{small black dog} \end{aligned}$
pdf-dc99cf88_E00043	015	■ 0.715	$\text{dog} \leq \text{black dog} \quad \rightsquigarrow \quad \text{dog} \stackrel{\pi(\text{black} \text{dog})}{\leq} \text{black dog}$	$\text{dog} \leq \text{black dog} \quad \rightsquigarrow \quad \text{dog} \stackrel{\pi(\text{black} \text{dog})}{\leq} \text{black dog}$	$\text{dog} \leq \text{black dog} \quad \rightsquigarrow \quad \text{dog} \stackrel{\pi(\text{black} \text{dog})}{\leq} \text{black dog}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
pdf-dc99cf88_E00034	013	■ 0.731	$\begin{aligned} & \backslash begin{aligned} \backslash rho_{\mathrm{dog}} \&= \pi(\backslash text{ small black } \backslash mid \backslash operatorname{dog}) \backslash rho_{\backslash text{small black dog}} \backslash \& + \pi(\backslash text{ small white } \backslash mid \backslash operatorname{dog}) \backslash rho_{\backslash text{small white dog}} \backslash \& + \pi(\backslash text{ big black } \backslash mid \backslash operatorname{dog}) \backslash rho_{\backslash text{big black dog}} \backslash \& = \frac{1}{3} \left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix} \right] + \frac{1}{3} \left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix} \right] + \frac{1}{3} \left[\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix} \right] \end{aligned}$	$\begin{aligned} \rho_{\text{dog}} &= \pi(\text{ small black } \mid \text{dog}) \rho_{\text{small black dog}} \\ &+ \pi(\text{ small white } \mid \text{dog}) \rho_{\text{small white dog}} \\ &+ \pi(\text{ big black } \mid \text{dog}) \rho_{\text{big black dog}} \\ &= \frac{1}{3} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \end{aligned}$	$\begin{aligned} \rho_{\text{dog}} &= \pi(\text{small black} \mid \text{dog}) \rho_{\text{small black dog}} \\ &+ \pi(\text{small white} \mid \text{dog}) \rho_{\text{small white dog}} \\ &+ \pi(\text{big black} \mid \text{dog}) \rho_{\text{big black dog}} \\ &= \frac{1}{3} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}. \end{aligned}$
pdf-dc99cf88_E00015	008	■ 0.809	$\hat{\rho}_{\backslash rho_{\backslash x_{i_k}} \cdots x_{i_1}} \backslash cdots x_{i_1} := \sum_{\backslash substack{i_{N-1}, \ldots, i_{k+1}} \backslash \backslash a, b} \psi_{i_{N-1} \cdots i_1 a} \bar{\psi}_{i_{N-1} \cdots i_1 b} x_a x_b^* .$	$\hat{\rho}_{x_{i_k} \cdots x_{i_1}} := \sum_{i_{N-1}, \dots, i_{k+1} \atop a, b} \psi_{i_{N-1} \cdots i_1 a} \bar{\psi}_{i_{N-1} \cdots i_1 b} x_a x_b^* .$	$\hat{\rho}_{x_{i_k} \cdots x_{i_1}} := \sum_{i_{N-1}, \dots, i_{k+1} \atop a, b} \psi_{i_{N-1} \cdots i_1 a} \bar{\psi}_{i_{N-1} \cdots i_1 b} x_a x_b^* .$
pdf-dc99cf88_E00031	012	■ 0.903	$\backslash rho_{\backslash x_{i_1}} = \sum_{i_{N-1}, \ldots, i_2} \pi \left(x_{i_{N-1}} \cdots x_{i_2} \mid x_{i_1} \right) \rho_{x_{i_{N-1}} \cdots x_{i_2} x_{i_1}}$	$\rho_{x_{i_1}} = \sum_{i_{N-1}, \dots, i_2} \pi \left(x_{i_{N-1}} \cdots x_{i_2} \mid x_{i_1} \right) \rho_{x_{i_{N-1}} \cdots x_{i_2} x_{i_1}}$	$\rho_{x_{i_1}} = \sum_{i_{N-1}, \dots, i_2} \pi(x_{i_{N-1}} \cdots x_{i_2} \mid x_{i_1}) \rho_{x_{i_{N-1}} \cdots x_{i_2} x_{i_1}}$
pdf-dc99cf88_E00017	008	■ 0.949	$\begin{aligned} & \backslash begin{aligned} \backslash hat{\backslash rho}_{\backslash x_{i_k} \cdots x_{i_1}} \backslash cdots x_{i_1} \backslash \&= \sum_{\backslash substack{i_{N-1}, \ldots, i_{k+1}} \backslash \backslash a, b} \psi_{i_{N-1} \cdots i_1 a} \bar{\psi}_{i_{N-1} \cdots i_1 b} x_a x_b^* \backslash \& + \sum_{i_{k+1}} \left(\sum_{i_{N-1}, \dots, i_{k+2} \atop a, b} \psi_{i_{N-1} \cdots i_1 a} \bar{\psi}_{i_{N-1} \cdots i_1 b} x_a x_b^* \right) \\ &= \sum_{i_{k+1}} \hat{\rho}_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}} . \end{aligned}$	$\begin{aligned} \hat{\rho}_{x_{i_k} \cdots x_{i_1}} &= \sum_{i_{N-1}, \dots, i_{k+1} \atop a, b} \psi_{i_{N-1} \cdots i_1 a} \bar{\psi}_{i_{N-1} \cdots i_1 b} x_a x_b^* \\ &= \sum_{i_{k+1}} \left(\sum_{i_{N-1}, \dots, i_{k+2} \atop a, b} \psi_{i_{N-1} \cdots i_1 a} \bar{\psi}_{i_{N-1} \cdots i_1 b} x_a x_b^* \right) \\ &= \sum_{i_{k+1}} \hat{\rho}_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}} . \end{aligned}$	$\begin{aligned} \hat{\rho}_{x_{i_k} \cdots x_{i_1}} &= \sum_{i_{N-1}, \dots, i_{k+1} \atop a, b} \psi_{i_{N-1} \cdots i_1 a} \bar{\psi}_{i_{N-1} \cdots i_1 b} x_a x_b^* \\ &= \sum_{i_{k+1}} \left(\sum_{i_{N-1}, \dots, i_{k+2} \atop a, b} \psi_{i_{N-1} \cdots i_1 a} \bar{\psi}_{i_{N-1} \cdots i_1 b} x_a x_b^* \right) \\ &= \sum_{i_{k+1}} \hat{\rho}_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}} . \end{aligned}$
pdf-dc99cf88_E00016	008	■ 0.953	$\backslash hat{\backslash rho}_{\backslash x_{i_k} \cdots x_{i_1}} \backslash cdots x_{i_1} := \sum_{i_{k+1}} \hat{\rho}_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}}$	$\hat{\rho}_{x_{i_k} \cdots x_{i_1}} = \sum_{i_{k+1}} \hat{\rho}_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}}$	$\hat{\rho}_{x_{i_k} \cdots x_{i_1}} = \sum_{i_{k+1}} \hat{\rho}_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}}$
pdf-dc99cf88_E00038	014	■ 0.974	$\backslash rho_{\mathrm{dog}} \geq \pi(\backslash text{ black } \backslash mid \backslash operatorname{dog}) \backslash rho_{\backslash text{black dog}} \geq \pi(\backslash text{ small black } \backslash mid \backslash operatorname{dog}) \backslash rho_{\backslash text{small black dog}} .$	$\rho_{\text{dog}} \geq \pi(\text{ black } \mid \text{dog}) \rho_{\text{black dog}} \geq \pi(\text{ small black } \mid \text{dog}) \rho_{\text{small black dog}} .$	$\rho_{\text{dog}} \geq \pi(\text{black} \mid \text{dog}) \rho_{\text{black dog}} \geq \pi(\text{small black} \mid \text{dog}) \rho_{\text{small black dog}} .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					

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pdf-dc99cf88_E0003 (3)	005	■ 0.990	$\operatorname{Pr}_{\psi}(\psi_i = \psi \mid x_i = a, y_i = b) = \frac{\sum_{j,b} \psi_{ij} x_{ij} y_{ij}}{\sum_{j,b} \psi_{ij} x_{ij} y_{ij}} = \sum_{j,b} \psi_{ij} x_{ij} y_{ij}.$	$\Pr_{\psi} = \psi \psi^* = \left(\sum_{i,a} \psi_{ia} x_{ia} y_a \right) \left(\sum_{j,b} \bar{\psi}_{jb} x_j^* y_b^* \right) = \sum_{i,a} \psi_{ia} \bar{\psi}_{jb} x_i x_j^* y_a y_b^*.$	$\Pr_{\psi} = \psi \psi^* = \left(\sum_{i,a} \psi_{ia} x_{ia} y_a \right) \left(\sum_{j,b} \bar{\psi}_{jb} x_j^* y_b^* \right) = \sum_{i,a} \psi_{ia} \bar{\psi}_{jb} x_i x_j^* y_a y_b^*.$
pdf-dc99cf88_E00033	012	■ 0.992	$\frac{1}{3} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = \rho_{\text{dog}} = \pi(\text{black} \mid \text{dog}) \rho_{\text{black dog}} + \pi(\text{white} \mid \text{dog}) \rho_{\text{white dog}}$	$\frac{1}{3} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = \rho_{\text{dog}} = \pi(\text{black} \mid \text{dog}) \rho_{\text{black dog}} + \pi(\text{white} \mid \text{dog}) \rho_{\text{white dog}}$	$\frac{1}{3} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = \rho_{\text{dog}} = \pi(\text{black} \mid \text{dog}) \rho_{\text{black dog}} + \pi(\text{white} \mid \text{dog}) \rho_{\text{white dog}}$
pdf-dc99cf88_E00030	012	■ 0.992	$\begin{aligned} \rho_{x_{i_k} \cdots x_{i_1}} &= \sum_{i_{k+1}} \pi(x_{i_{k+1}} \mid x_{i_k} \cdots x_{i_1}) \rho_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}} \\ &= \sum_{i_{k+2}, i_{k+1}} \pi(x_{i_{k+2}} x_{i_{k+1}} \mid x_{i_k} \cdots x_{i_1}) \rho_{x_{i_{k+2}} x_{i_{k+1}} x_{i_k} \cdots x_{i_1}} \\ &= \vdots \\ &= \sum_{i_{N-1}, \dots, i_{k+1}} \pi(x_{i_{N-1}} \cdots x_{i_{k+1}} \mid x_{i_k} \cdots x_{i_1}) \rho_{x_{i_{N-1}} \cdots x_{i_k} \cdots x_{i_1}} \end{aligned}$	$\rho_{x_{i_k} \cdots x_{i_1}} = \sum_{i_{k+1}} \pi(x_{i_{k+1}} \mid x_{i_k} \cdots x_{i_1}) \rho_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}} \\ = \sum_{i_{k+2}, i_{k+1}} \pi(x_{i_{k+2}} x_{i_{k+1}} \mid x_{i_k} \cdots x_{i_1}) \rho_{x_{i_{k+2}} x_{i_{k+1}} x_{i_k} \cdots x_{i_1}} \\ = \vdots \\ = \sum_{i_{N-1}, \dots, i_{k+1}} \pi(x_{i_{N-1}} \cdots x_{i_{k+1}} \mid x_{i_k} \cdots x_{i_1}) \rho_{x_{i_{N-1}} \cdots x_{i_k} \cdots x_{i_1}}$	$\rho_{x_{i_k} \cdots x_{i_1}} = \sum_{i_{k+1}} \pi(x_{i_{k+1}} \mid x_{i_k} \cdots x_{i_1}) \rho_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}} \\ = \sum_{i_{k+2}, i_{k+1}} \pi(x_{i_{k+2}} x_{i_{k+1}} \mid x_{i_k} \cdots x_{i_1}) \rho_{x_{i_{k+2}} x_{i_{k+1}} x_{i_k} \cdots x_{i_1}} \\ = \vdots \\ = \sum_{i_{N-1}, \dots, i_{k+1}} \pi(x_{i_{N-1}} \cdots x_{i_{k+1}} \mid x_{i_k} \cdots x_{i_1}) \rho_{x_{i_{N-1}} \cdots x_{i_k} \cdots x_{i_1}}$
pdf-dc99cf88_E00028	011	■ 0.993	$\begin{aligned} \rho_{x_{i_k} \cdots x_{i_1}} &= \frac{\hat{\rho}_{x_{i_k} \cdots x_{i_1}}}{\pi(x_{i_k} \cdots x_{i_1})} = \sum_{i_{k+1}} \frac{\pi(x_{i_{k+1}} x_{i_k} \cdots x_{i_1})}{\pi(x_{i_k} \cdots x_{i_1})} \frac{\hat{\rho}_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}}}{\pi(x_{i_{k+1}} x_{i_k} \cdots x_{i_1})} \\ &= \sum_{i_{k+1}} \pi(x_{i_{k+1}} \mid x_{i_k} \cdots x_{i_1}) \rho_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}}. \end{aligned}$	$\rho_{x_{i_k} \cdots x_{i_1}} = \frac{\hat{\rho}_{x_{i_k} \cdots x_{i_1}}}{\pi(x_{i_k} \cdots x_{i_1})} = \sum_{i_{k+1}} \frac{\pi(x_{i_{k+1}} x_{i_k} \cdots x_{i_1})}{\pi(x_{i_k} \cdots x_{i_1})} \frac{\hat{\rho}_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}}}{\pi(x_{i_{k+1}} x_{i_k} \cdots x_{i_1})} \\ = \sum_{i_{k+1}} \pi(x_{i_{k+1}} \mid x_{i_k} \cdots x_{i_1}) \rho_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}}.$	$\rho_{x_{i_k} \cdots x_{i_1}} = \frac{\hat{\rho}_{x_{i_k} \cdots x_{i_1}}}{\pi(x_{i_k} \cdots x_{i_1})} = \sum_{i_{k+1}} \frac{\pi(x_{i_{k+1}} x_{i_k} \cdots x_{i_1})}{\pi(x_{i_k} \cdots x_{i_1})} \frac{\hat{\rho}_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}}}{\pi(x_{i_{k+1}} x_{i_k} \cdots x_{i_1})} \\ = \sum_{i_{k+1}} \pi(x_{i_{k+1}} \mid x_{i_k} \cdots x_{i_1}) \rho_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}}.$
pdf-dc99cf88_E00036 (10)	013	■ 0.995	$\hat{\rho}_{\text{dog}} \geq \hat{\rho}_{\text{black dog}} \geq \hat{\rho}_{\text{small black dog}}.$	$\hat{\rho}_{\text{dog}} \geq \hat{\rho}_{\text{black dog}} \geq \hat{\rho}_{\text{small black dog}}.$	$\hat{\rho}_{\text{dog}} \geq \hat{\rho}_{\text{black dog}} \geq \hat{\rho}_{\text{small black dog}}.$
pdf-dc99cf88_E00046	016	■ 0.996	$\mathcal{D}(\hat{\rho}_s, \hat{\rho}_{s'}) = \begin{cases} \operatorname{tr} \hat{\rho}_{s'} / \operatorname{tr} \hat{\rho}_s & \text{if } s \leq s' \\ 0 & \text{otherwise} \end{cases}.$	$\mathcal{D}(\hat{\rho}_s, \hat{\rho}_{s'}) = \begin{cases} \operatorname{tr} \hat{\rho}_{s'} / \operatorname{tr} \hat{\rho}_s & \text{if } s \leq s' \\ 0 & \text{otherwise} \end{cases}.$	$\mathcal{D}(\hat{\rho}_s, \hat{\rho}_{s'}) = \begin{cases} \operatorname{tr} \hat{\rho}_{s'} / \operatorname{tr} \hat{\rho}_s & \text{if } s \leq s' \\ 0 & \text{otherwise} \end{cases}.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
pdf-dc99cf88_EQ0019	009	■ 0.997	$\psi=\frac{1}{\sqrt{5}}\left(\begin{array}{cccccc}\text{small}& \otimes & \text{black}& \otimes & \text{dog}& \otimes & \text{barks}& +\\ \text{small}& \otimes & \text{white}& \otimes & \text{dog}& \otimes & \text{barks}& +\\ \text{big}& \otimes & \text{black}& \otimes & \text{dog}& \otimes & \text{runs}& +\\ \text{big}& \otimes & \text{white}& \otimes & \text{cat}& \otimes & \text{runs}& +\\ \text{small}& \otimes & \text{black}& \otimes & \text{cat}& \otimes & \text{runs}& \end{array}\right)$	$\psi=\frac{1}{\sqrt{5}}\left(\begin{array}{cccccc}\text{small}& \otimes & \text{black}& \otimes & \text{dog}& \otimes & \text{barks}& +\\ \text{small}& \otimes & \text{white}& \otimes & \text{dog}& \otimes & \text{barks}& +\\ \text{big}& \otimes & \text{black}& \otimes & \text{dog}& \otimes & \text{runs}& +\\ \text{big}& \otimes & \text{white}& \otimes & \text{cat}& \otimes & \text{runs}& +\\ \text{small}& \otimes & \text{black}& \otimes & \text{cat}& \otimes & \text{runs}& \end{array}\right)$	
pdf-dc99cf88_EQ0027	011	■ 0.997	$\rho_{x_{i_k}\cdots x_{i_1}}\mid x_{i_k}\cdots x_{i_1}=\sum_{i_{k+1}}\pi\left(x_{i_{k+1}}\mid x_{i_k}\cdots x_{i_1}\right)\rho_{x_{i_{k+1}}x_{i_k}\cdots x_{i_1}}.$	$\rho_{x_{i_k}\cdots x_{i_1}}=\sum_{i_{k+1}}\pi\left(x_{i_{k+1}}\mid x_{i_k}\cdots x_{i_1}\right)\rho_{x_{i_{k+1}}x_{i_k}\cdots x_{i_1}}.$	$\rho_{x_{i_k}\cdots x_{i_1}}=\sum_{i_{k+1}}\pi\left(x_{i_{k+1}}\mid x_{i_k}\cdots x_{i_1}\right)\rho_{x_{i_{k+1}}x_{i_k}\cdots x_{i_1}}.$
pdf-dc99cf88_EQ0022	010	■ 0.998	$\hat{\rho}_{\text{black dog}}=\frac{1}{5}\left[\begin{array}{cc}1&0\\0&1\end{array}\right]\quad\hat{\rho}_{\text{small black dog}}=\frac{1}{5}\left[\begin{array}{cc}1&0\\0&0\end{array}\right]$	$\hat{\rho}_{\text{black dog}}=\frac{1}{5}\left[\begin{array}{cc}1&0\\0&1\end{array}\right]\quad\hat{\rho}_{\text{small black dog}}=\frac{1}{5}\left[\begin{array}{cc}1&0\\0&0\end{array}\right]$	$\hat{\rho}_{\text{black dog}}=\frac{1}{5}\left[\begin{array}{cc}1&0\\0&1\end{array}\right]\quad\hat{\rho}_{\text{small black dog}}=\frac{1}{5}\left[\begin{array}{cc}1&0\\0&0\end{array}\right]$
pdf-dc99cf88_EQ0035 (9)	013	■ 0.998	$\hat{\rho}_{x_{i_k}\cdots x_{i_1}}\geq\hat{\rho}_{x_{i_{k+1}}x_{i_k}\cdots x_{i_1}}$	$\hat{\rho}_{x_{i_k}\cdots x_{i_1}}\geq\hat{\rho}_{x_{i_{k+1}}x_{i_k}\cdots x_{i_1}}$	$\hat{\rho}_{x_{i_k}\cdots x_{i_1}}\geq\hat{\rho}_{x_{i_{k+1}}x_{i_k}\cdots x_{i_1}}$
pdf-dc99cf88_EQ0023	011	■ 0.999	$\sum_{i_{N-1},\ldots,i_{k+1},a}\left \left\langle\psi,x_{i_{N-1}}\cdots x_{i_3}x_{i_2}x_{i_1}x_a\right\rangle\right ^2=\sum_{i_{N-1},\ldots,i_{k+1},a}\pi\left(x_{i_{N-1}},\ldots,x_{i_3},x_{i_2},x_{i_1},x_a\right)$	$\sum_{i_{N-1},\ldots,i_{k+1},a}\left \left\langle\psi,x_{i_{N-1}}\cdots x_{i_3}x_{i_2}x_{i_1}x_a\right\rangle\right ^2=\sum_{i_{N-1},\ldots,i_{k+1},a}\pi\left(x_{i_{N-1}},\ldots,x_{i_3},x_{i_2},x_{i_1},x_a\right)$	$\sum_{i_{N-1},\ldots,i_{k+1},a}\left \left\langle\psi,x_{i_{N-1}}\cdots x_{i_3}x_{i_2}x_{i_1}x_a\right\rangle\right ^2=\sum_{i_{N-1},\ldots,i_{k+1},a}\pi\left(x_{i_{N-1}},\ldots,x_{i_3},x_{i_2},x_{i_1},x_a\right)$
pdf-dc99cf88_EQ0010	006	■ 0.999	$\rho_X=\frac{1}{4}\left[\begin{array}{cccc}1&1&0&0\\1&1&0&0\\0&0&1&1\\0&0&1&1\end{array}\right]\quad\rho_Y=\frac{1}{4}\left[\begin{array}{cc}2&0\\0&2\end{array}\right]$	$\rho_X=\frac{1}{4}\left[\begin{array}{cccc}1&1&0&0\\1&1&0&0\\0&0&1&1\\0&0&1&1\end{array}\right]\quad\rho_Y=\frac{1}{4}\left[\begin{array}{cc}2&0\\0&2\end{array}\right]$	$\rho_X=\frac{1}{4}\left[\begin{array}{cccc}1&1&0&0\\1&1&0&0\\0&0&1&1\\0&0&1&1\end{array}\right]\quad\rho_Y=\frac{1}{4}\left[\begin{array}{cc}2&0\\0&2\end{array}\right]$
pdf-dc99cf88_EQ0014	007	■ 0.999	$\hat{\rho}_{x_{i_2}x_{i_1}}:=\sum_{i_4,i_3,a,b}\psi_{i_4i_3i_2i_1a}\bar{\psi}_{i_4i_3i_2i_1b}x_a^*x_b$	$\hat{\rho}_{x_{i_2}x_{i_1}}:=\sum_{i_4,i_3,a,b}\psi_{i_4i_3i_2i_1a}\bar{\psi}_{i_4i_3i_2i_1b}x_a^*x_b$	$\hat{\rho}_{x_{i_2}x_{i_1}}:=\sum_{i_4,i_3,a,b}\psi_{i_4i_3i_2i_1a}\bar{\psi}_{i_4i_3i_2i_1b}x_a^*x_b$
pdf-dc99cf88_EQ0005	005	■ 1.000	$\rho_Y=\left[\begin{array}{cccc}\pi_Y\left(y_1\right)&&&*\\&\pi_Y\left(y_2\right)&&\\&&\ddots&*&&&\pi_Y\left(y_m\right)\end{array}\right]$	$\rho_Y=\left[\begin{array}{cccc}\pi_Y\left(y_1\right)&&&*\\&\pi_Y\left(y_2\right)&&\\&&\ddots&*&&&\pi_Y\left(y_m\right)\end{array}\right]$	$\rho_Y=\left[\begin{array}{cccc}\pi_Y\left(y_1\right)&&&*\\&\pi_Y\left(y_2\right)&&\\&&\ddots&*&&&\pi_Y\left(y_m\right)\end{array}\right]$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
pdf-dc99cf88_E00018 (7)	009	■ 1.000	$\hat{\rho}_{x_{i_1}} = \sum_{i_2} \hat{\rho}_{x_{i_2}} \hat{\rho}_{x_{i_1}} = \sum_{i_3, i_2} \hat{\rho}_{x_{i_3} x_{i_2} x_{i_1}} = \cdots = \sum_{i_{N-1}, \dots, i_3, i_2} \hat{\rho}_{x_{i_{N-1}} \cdots x_{i_3} x_{i_2} x_{i_1}}$		
pdf-dc99cf88_E00026	011	■ 1.000	$\hat{\rho}_{\mathrm{dog}} = \frac{1}{5} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$	$\hat{\rho}_{\mathrm{dog}} = \frac{1}{5} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \quad \rho_{\mathrm{dog}} = \frac{1}{3} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$	$\hat{\rho}_{\mathrm{dog}} = \frac{1}{5} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \quad \rho_{\mathrm{dog}} = \frac{1}{3} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$
pdf-dc99cf88_E00040	014	■ 1.000	$(x_{i_{N-1}}, \dots, \overbrace{x_{i_k}, \dots, x_{i_1}}^s, x_a)$	$(x_{i_{N-1}}, \dots, \overbrace{x_{i_k}, \dots, x_{i_1}}^s, x_a)$	$(x_{i_{N-1}}, \dots, \overbrace{x_{i_k}, \dots, x_{i_1}}^s, x_a)$
pdf-dc99cf88_E00001 (1)	004	■ 1.000	$\psi = \sum_{s \in S} \sqrt{\pi(s)} s$	$\psi = \sum_{s \in S} \sqrt{\pi(s)} s$	$\psi = \sum_{s \in S} \sqrt{\pi(s)} s$
pdf-dc99cf88_E00002 (2)	005	■ 1.000	$\psi = \sum_{i,a} \psi_{ia} x_i y_a$	$\psi = \sum_{i,a} \psi_{ia} x_i y_a$	$\psi = \sum_{i,a} \psi_{ia} x_i y_a$
pdf-dc99cf88_E00004 (4)	005	■ 1.000	$\rho_Y = \mathrm{tr}_X \mathrm{Pr}_\psi = \sum_{i,a} \psi_{ia} \bar{\psi}_{jb} \mathrm{tr}_X(x_i x_j^* y_a y_b^*) = \sum_{i,a} \psi_{ia} \bar{\psi}_{jb} \mathrm{tr}(x_i x_j^*) \cdot y_a y_b^* = \sum_{a,b} \psi_{ia} \bar{\psi}_{ib} y_a y_b^*$	$\rho_Y = \mathrm{tr}_X \mathrm{Pr}_\psi = \sum_{i,a} \psi_{ia} \bar{\psi}_{jb} \mathrm{tr}_X(x_i x_j^* y_a y_b^*) = \sum_{i,a} \psi_{ia} \bar{\psi}_{jb} \mathrm{tr}(x_i x_j^*) \cdot y_a y_b^* = \sum_{a,b} \psi_{ia} \bar{\psi}_{ib} y_a y_b^*$	$\rho_Y = \mathrm{tr}_X \mathrm{Pr}_\psi = \sum_{i,a} \psi_{ia} \bar{\psi}_{jb} \mathrm{tr}_X(x_i x_j^* y_a y_b^*) = \sum_{i,a} \psi_{ia} \bar{\psi}_{jb} \mathrm{tr}(x_i x_j^*) \cdot y_a y_b^* = \sum_{a,b} \psi_{ia} \bar{\psi}_{ib} y_a y_b^*$
pdf-dc99cf88_E00006	005	■ 1.000	$\sum_i \sqrt{\pi(x_i y_a) \pi(x_i y_b)} = \frac{1}{\sqrt{\pi_Y(y_a) \pi_Y(y_b)}} \sum_i \sqrt{\pi(x_i, y_a) \pi(x_i, y_b)} = \frac{(\rho_Y)_{ab}}{\sqrt{\pi_Y(y_a) \pi_Y(y_b)}}$	$\sum_i \sqrt{\pi(x_i y_a) \pi(x_i y_b)} = \frac{1}{\sqrt{\pi_Y(y_a) \pi_Y(y_b)}} \sum_i \sqrt{\pi(x_i, y_a) \pi(x_i, y_b)} = \frac{(\rho_Y)_{ab}}{\sqrt{\pi_Y(y_a) \pi_Y(y_b)}}$	$\sum_i \sqrt{\pi(x_i y_a) \pi(x_i y_b)} = \frac{1}{\sqrt{\pi_Y(y_a) \pi_Y(y_b)}} \sum_i \sqrt{\pi(x_i, y_a) \pi(x_i, y_b)} = \frac{(\rho_Y)_{ab}}{\sqrt{\pi_Y(y_a) \pi_Y(y_b)}}$
pdf-dc99cf88_E00007	006	■ 1.000	$\rho_Y(y_c) = \sum_{i,a} \sqrt{\pi(x_i, y_a) \pi(x_i, y_c)} y_a = \sum_{i,a} \sqrt{\pi(x_i, y_a) \pi(x_i, y_d)} y_a = \rho_Y(y_d)$	$\rho_Y(y_c) = \sum_{i,a} \sqrt{\pi(x_i, y_a) \pi(x_i, y_c)} y_a = \sum_{i,a} \sqrt{\pi(x_i, y_a) \pi(x_i, y_d)} y_a = \rho_Y(y_d)$	$\rho_Y(y_c) = \sum_{i,a} \sqrt{\pi(x_i, y_a) \pi(x_i, y_c)} y_a = \sum_{i,a} \sqrt{\pi(x_i, y_a) \pi(x_i, y_d)} y_a = \rho_Y(y_d)$
pdf-dc99cf88_E00011 (5)	007	■ 1.000	$\psi = \sum_{i_{N-1}, \dots, i_1, a} \psi_{i_{N-1} \cdots i_1 a} x_{i_{N-1}} \cdots x_{i_1} x_a$	$\psi = \sum_{i_{N-1}, \dots, i_1, a} \psi_{i_{N-1} \cdots i_1 a} x_{i_{N-1}} \cdots x_{i_1} x_a$	$\psi = \sum_{i_{N-1}, \dots, i_1, a} \psi_{i_{N-1} \cdots i_1 a} x_{i_{N-1}} \cdots x_{i_1} x_a$
pdf-dc99cf88_E00012	007	■ 1.000	$\mathrm{Pr}_\psi = \psi \psi^* = \sum_{i_{N-1}, \dots, i_1, a} \psi_{i_{N-1} \cdots i_1 a} \bar{\psi}_{j_{N-1} \cdots j_1 b} x_{i_{N-1}} \cdots x_{i_1} x_a x_{j_{N-1}}^* \cdots x_{j_1}^* x_b^*$	$\mathrm{Pr}_\psi = \psi \psi^* = \sum_{i_{N-1}, \dots, i_1, a} \psi_{i_{N-1} \cdots i_1 a} \bar{\psi}_{j_{N-1} \cdots j_1 b} x_{i_{N-1}} \cdots x_{i_1} x_a x_{j_{N-1}}^* \cdots x_{j_1}^* x_b^*$	$\mathrm{Pr}_\psi = \psi \psi^* = \sum_{i_{N-1}, \dots, i_1, a} \psi_{i_{N-1} \cdots i_1 a} \bar{\psi}_{j_{N-1} \cdots j_1 b} x_{i_{N-1}} \cdots x_{i_1} x_a x_{j_{N-1}}^* \cdots x_{j_1}^* x_b^*$
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Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
pdf-dc99cf88_EQ0013 (6)	007	■ 1.000	$\rho_V=\sum_{i=1}^N a_i b_i \psi_{i=1}^N x_i^*$	$\rho_V = \sum_{i=1, \dots, i_1 a, b} \psi_{i_{N-1} \dots i_1 a} \bar{\psi}_{i_{N-1} \dots i_1 b} x_a x_b^*$	$\rho_V = \sum_{i_{N-1}, \dots, i_1 a, b} \psi_{i_{N-1} \dots i_1 a} \bar{\psi}_{i_{N-1} \dots i_1 b} x_a x_b^*$
pdf-dc99cf88_EQ0024	011	■ 1.000	$\rho_{x_{i_k} \dots x_{i_1}} := \hat{\rho}_{x_{i_k} \dots x_{i_1}} / \pi(x_{i_k} \dots x_{i_1})$	$\rho_{x_{i_k} \dots x_{i_1}} := \hat{\rho}_{x_{i_k} \dots x_{i_1}} / \pi(x_{i_k} \dots x_{i_1})$	$\rho_{x_{i_k} \dots x_{i_1}} := \hat{\rho}_{x_{i_k} \dots x_{i_1}} / \pi(x_{i_k} \dots x_{i_1})$
pdf-dc99cf88_EQ0025	011	■ 1.000	$\frac{1}{\pi(x_{i_k} \dots x_{i_1})} \sum_{i_{N-1}, \dots, i_{k+1}} \langle \psi, x_{i_{N-1}} \dots x_{i_1} x_a \rangle ^2 = \frac{\pi(x_{i_k} \dots x_{i_1}, x_a)}{\pi(x_{i_k} \dots x_{i_1})} = \pi(x_a x_{i_k} \dots x_{i_1})$	$\frac{1}{\pi(x_{i_k} \dots x_{i_1})} \sum_{i_{N-1}, \dots, i_{k+1}} \langle \psi, x_{i_{N-1}} \dots x_{i_1} x_a \rangle ^2 = \frac{\pi(x_{i_k} \dots x_{i_1}, x_a)}{\pi(x_{i_k} \dots x_{i_1})} = \pi(x_a x_{i_k} \dots x_{i_1})$	$\frac{1}{\pi(x_{i_k} \dots x_{i_1})} \sum_{i_{N-1}, \dots, i_{k+1}} \langle \psi, x_{i_{N-1}} \dots x_{i_1} x_a \rangle ^2 = \frac{\pi(x_{i_k} \dots x_{i_1}, x_a)}{\pi(x_{i_k} \dots x_{i_1})} = \pi(x_a x_{i_k} \dots x_{i_1})$
pdf-dc99cf88_EQ0029 (8)	012	■ 1.000	$\rho_{x_{i_1}} = \sum_{i_{N-1}, \dots, i_2} \pi(x_{i_{N-1}} \dots x_{i_2} x_{i_1}) \rho_{x_{i_{N-1}} \dots x_{i_2} x_{i_1}}$	$\rho_{x_{i_1}} = \sum_{i_{N-1}, \dots, i_2} \pi(x_{i_{N-1}} \dots x_{i_2} x_{i_1}) \rho_{x_{i_{N-1}} \dots x_{i_2} x_{i_1}}$	$\rho_{x_{i_1}} = \sum_{i_{N-1}, \dots, i_2} \pi(x_{i_{N-1}} \dots x_{i_2} x_{i_1}) \rho_{x_{i_{N-1}} \dots x_{i_2} x_{i_1}}$
pdf-dc99cf88_EQ0032	012	■ 1.000	$\rho_{\text{black dog}} = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \rho_{\text{white dog}} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	$\rho_{\text{black dog}} = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \rho_{\text{white dog}} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	$\rho_{\text{black dog}} = \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \rho_{\text{white dog}} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$
pdf-dc99cf88_EQ0037 (11)	014	■ 1.000	$\rho_{x_{i_k} \dots x_{i_1}} \geq \pi(x_{i_{k+1}} x_{i_k} \dots x_{i_1}) \rho_{x_{i_{k+1}} x_{i_k} \dots x_{i_1}}$	$\rho_{x_{i_k} \dots x_{i_1}} \geq \pi(x_{i_{k+1}} x_{i_k} \dots x_{i_1}) \rho_{x_{i_{k+1}} x_{i_k} \dots x_{i_1}}$	$\rho_{x_{i_k} \dots x_{i_1}} \geq \pi(x_{i_{k+1}} x_{i_k} \dots x_{i_1}) \rho_{x_{i_{k+1}} x_{i_k} \dots x_{i_1}}$
pdf-dc99cf88_EQ0041	015	■ 1.000	$\left\{ x_{i_2} \text{ dog} \mid x_{i_2} \in X \right\} \sqcup \left\{ x_{i_3} x_{i_2} \text{ dog} \mid x_{i_3}, x_{i_2} \in X \right\} \sqcup \dots \sqcup \left\{ x_{i_{N-1}} \dots x_{i_3} x_{i_2} \text{ dog} \mid x_{i_{N-1}}, \dots, x_{i_3}, x_{i_2} \in X \right\}$	$\{x_{i_2} \text{ dog} \mid x_{i_2} \in X\} \sqcup \{x_{i_3} x_{i_2} \text{ dog} \mid x_{i_3}, x_{i_2} \in X\} \sqcup \dots \sqcup \{x_{i_{N-1}} \dots x_{i_3} x_{i_2} \text{ dog} \mid x_{i_{N-1}}, \dots, x_{i_3}, x_{i_2} \in X\}$	$\{x_{i_2} \text{ dog} \mid x_{i_2} \in X\} \sqcup \{x_{i_3} x_{i_2} \text{ dog} \mid x_{i_3}, x_{i_2} \in X\} \sqcup \dots \sqcup \{x_{i_{N-1}} \dots x_{i_3} x_{i_2} \text{ dog} \mid x_{i_{N-1}}, \dots, x_{i_3}, x_{i_2} \in X\}$
pdf-dc99cf88_EQ0044	016	■ 1.000	$\mathcal{C}(c, c') \leq \mathcal{D}(fc, fc')$	$\mathcal{C}(c, c') \leq \mathcal{D}(fc, fc')$	$\mathcal{C}(c, c') \leq \mathcal{D}(fc, fc')$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value					

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered
pdf-dc99cf88_F00001	001	—	$\{ \ }^{+}$	$+$
pdf-dc99cf88_F00002	003	—	s	s
pdf-dc99cf88_F00003	003	—	$\rho_{\scriptscriptstyle s}$	ρ_s
pdf-dc99cf88_F00004	003	—	$s \mapsto \rho_{\scriptscriptstyle s}$	$s \mapsto \rho_s$
pdf-dc99cf88_F00005	003	—	$\leq$	$\leq$
pdf-dc99cf88_F00006	003	—	$\hat{\rho}_{\scriptscriptstyle s}$	$\hat{\rho}_s$
pdf-dc99cf88_F00007	003	—	$s \mapsto \hat{\rho}_{\scriptscriptstyle s}$	$s \mapsto \hat{\rho}_s$
pdf-dc99cf88_F00008	004	—	S	S
pdf-dc99cf88_F00009	004	—	$V=\mathbb{C}^{\scriptscriptstyle S}$	$V = \mathbb{C}^S$
pdf-dc99cf88_F00010	004	—	V	V
pdf-dc99cf88_F00011	004	—	$s \in S$	$s \in S$
pdf-dc99cf88_F00012	004	—	$s \in V$	$s \in V$
pdf-dc99cf88_F00013	004	—	$S=\left\{s_{\scriptscriptstyle 1}\right\}, \ldots, s_{\scriptscriptstyle d}\right\}$	$S = \{s_1, \dots, s_d\}$
pdf-dc99cf88_F00014	004	—	$s_{\scriptscriptstyle i}$	s_i
pdf-dc99cf88_F00015	004	—	i	i
pdf-dc99cf88_F00016	004	—	$\mathbb{C}^{\scriptscriptstyle d}$	$\mathbb{C}^d$
pdf-dc99cf88_F00017	004	—	$V \cong \mathbb{C}^{\scriptscriptstyle d}$	$V \cong \mathbb{C}^d$
pdf-dc99cf88_F00018	004	—	$\left\langle s_{\scriptscriptstyle i}, s_{\scriptscriptstyle j}\right\rangle$	$\langle s_i, s_j \rangle$
pdf-dc99cf88_F00019	004	—	$i=j$	$i = j$
pdf-dc99cf88_F00020	004	—	$v \in V$	$v \in V$
pdf-dc99cf88_F00021	004	—	$v^{\scriptscriptstyle *}: V \rightarrow \mathbb{C}$	$v^*: V \rightarrow \mathbb{C}$
pdf-dc99cf88_F00022	004	—	$v^{\scriptscriptstyle \prime} \mapsto \left\langle v, v^{\scriptscriptstyle \prime}\right\rangle$	$v' \mapsto \langle v, v' \rangle$
pdf-dc99cf88_F00023	004	—	$V^{\scriptscriptstyle *}:=\operatorname{hom}(V, \mathbb{C})$	$V^* := \operatorname{hom}(V, \mathbb{C})$
pdf-dc99cf88_F00024	004	—	W	W
pdf-dc99cf88_F00025	004	—	$V \otimes W$	$V \otimes W$
pdf-dc99cf88_F00026	004	—	vw	vw
pdf-dc99cf88_F00027	004	—	$v \otimes w$	$v \otimes w$
pdf-dc99cf88_F00028	004	—	$w v^{\scriptscriptstyle *}$	wv^*
pdf-dc99cf88_F00029	004	—	$W \otimes V^{\scriptscriptstyle *}$	$W \otimes V^*$
pdf-dc99cf88_F00030	004	—	$V \rightarrow W$	$V \rightarrow W$
pdf-dc99cf88_F00031	004	—	$v^{\scriptscriptstyle \prime} \mapsto w\left\langle v, v^{\scriptscriptstyle \prime}\right\rangle$	$v' \mapsto w \langle v, v' \rangle$
pdf-dc99cf88_F00032	004	—	$\operatorname{End}(V)$	$\operatorname{End}(V)$
pdf-dc99cf88_F00033	004	—	$\psi \in V$	$\psi \in V$
pdf-dc99cf88_F00034	004	—	$\psi \psi^{\scriptscriptstyle *} \in V \otimes V^{\scriptscriptstyle *}$	$\psi \psi^* \in V \otimes V^*$
pdf-dc99cf88_F00035	004	—	ψ	ψ
pdf-dc99cf88_F00036	004	—	$\operatorname{Pr}_{\scriptscriptstyle \psi}$	Pr_{ψ}
pdf-dc99cf88_F00037	004	—	ρ	ρ
pdf-dc99cf88_F00038	004	—	$\rho \geq 0$	$\rho \geq 0$
pdf-dc99cf88_F00039	004	—	$\pi_{\scriptscriptstyle \rho}: S \rightarrow \mathbb{R}$	$\pi_{\rho}: S \rightarrow \mathbb{R}$
pdf-dc99cf88_F00040	004	—	$\pi_{\scriptscriptstyle \rho}(s):=\left\langle \rho s, s\right\rangle$	$\pi_{\rho}(s) := \langle \rho s, s \rangle$
pdf-dc99cf88_F00041	004	—	$\pi: S \rightarrow \mathbb{R}$	$\pi: S \rightarrow \mathbb{R}$
pdf-dc99cf88_F00042	004	—	π	π
pdf-dc99cf88_F00043	004	—	$\sum_{\scriptscriptstyle s} \pi(s) s s^{\scriptscriptstyle *}$	$\sum_s \pi(s) s s^*$
pdf-dc99cf88_F00044	004	—	$\pi(s)$	$\pi(s)$
pdf-dc99cf88_F00045	004	—	$\operatorname{Pr}_{\scriptscriptstyle \psi}: V \rightarrow V$	$\operatorname{Pr}_{\psi}: V \rightarrow V$
pdf-dc99cf88_F00046	004	—	$\pi_{\scriptscriptstyle \operatorname{Pr}_{\scriptscriptstyle \psi}}(s)=\left\langle \operatorname{Pr}_{\scriptscriptstyle \psi} s, s\right\rangle=\sqrt{\pi(s)^2}=\pi(s)$	$\pi_{\operatorname{Pr}_{\psi}}(s) = \langle \operatorname{Pr}_{\psi} s, s \rangle = \sqrt{\pi(s)^2} = \pi(s)$
pdf-dc99cf88_F00047	004	—	$\pi \mapsto \operatorname{Pr}_{\scriptscriptstyle \psi}$	$\pi \mapsto \operatorname{Pr}_{\psi}$
pdf-dc99cf88_F00048	004	—	A	A
pdf-dc99cf88_F00049	004	—	B	B
pdf-dc99cf88_F00050	004	—	$\operatorname{End}(A \otimes B) \cong \operatorname{End}(A) \otimes$	$\operatorname{End}(A \otimes B) \cong \operatorname{End}(A) \otimes$
pdf-dc99cf88_F00051	004	—	$\operatorname{End}(B)$	$\operatorname{End}(B)$
pdf-dc99cf88_F00052	004	—	$\operatorname{tr}_{\scriptscriptstyle A}:=\operatorname{tr} \otimes \operatorname{id}_{\scriptscriptstyle B}$	$\operatorname{tr}_A := \operatorname{tr} \otimes \operatorname{id}_B$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
pdf-dc99cf88_F00053	004	—	$\operatorname{tr}_{\{B\}} := \operatorname{tr}_{\{A\}} \otimes \operatorname{tr}$	$\operatorname{tr}_B := \operatorname{id}_A \otimes \operatorname{tr}$
pdf-dc99cf88_F00054	004	—	$\rho: A \otimes B \rightarrow$	$\rho: A \otimes B \rightarrow$
pdf-dc99cf88_F00055	004	—	$A \otimes B$	$A \otimes B$
pdf-dc99cf88_F00056	004	—	$\rho_{\{B\}} := \operatorname{tr}_{\{A\}} \rho: B \rightarrow B$	$\rho_B := \operatorname{tr}_A \rho: B \rightarrow B$
pdf-dc99cf88_F00057	004	—	$\rho_{\{A\}} := \operatorname{tr}_{\{B\}} \rho: A \rightarrow A$	$\rho_A := \operatorname{tr}_B \rho: A \rightarrow A$
pdf-dc99cf88_F00058	004	—	$S = X \times Y$	$S = X \times Y$
pdf-dc99cf88_F00059	004	—	$X = \{x_1, \ldots, x_n\}$	$X = \{x_1, \ldots, x_n\}$
pdf-dc99cf88_F00060	004	—	$Y = \{y_1, \ldots, y_m\}$	$Y = \{y_1, \ldots, y_m\}$
pdf-dc99cf88_F00061	004	—	$\operatorname{Pr}_{\psi}: \mathbb{C}^X \otimes \mathbb{C}^Y \rightarrow \mathbb{C}^X \otimes \mathbb{C}^Y$	$\operatorname{Pr}_{\psi}: \mathbb{C}^X \otimes \mathbb{C}^Y \rightarrow \mathbb{C}^X \otimes \mathbb{C}^Y$
pdf-dc99cf88_F00062	005	—	$\psi_{ia} = \sqrt{\pi(x_i, y_a)}$	$\psi_{ia} = \sqrt{\pi(x_i, y_a)}$
pdf-dc99cf88_F00063	005	—	$\rho_Y: \mathbb{C}^Y \rightarrow \mathbb{C}^Y$	$\rho_Y: \mathbb{C}^Y \rightarrow \mathbb{C}^Y$
pdf-dc99cf88_F00064	005	—	$\operatorname{tr} (x_i x_j^*) = \langle x_j, x_i \rangle$	$\operatorname{tr} (x_i x_j^*) = \langle x_j, x_i \rangle$
pdf-dc99cf88_F00065	005	—	a	a
pdf-dc99cf88_F00066	005	—	ρ_Y	ρ_Y
pdf-dc99cf88_F00067	005	—	$\pi_Y(y_a) := \sum_i \psi_{ia} \overline{\psi_{ia}} = \sum_i \pi(x_i, y_a)$	$\pi_Y(y_a) := \sum_i \psi_{ia} \overline{\psi_{ia}} = \sum_i \pi(x_i, y_a)$
pdf-dc99cf88_F00068	005	—	$\pi_Y: Y \rightarrow \mathbb{R}$	$\pi_Y: Y \rightarrow \mathbb{R}$
pdf-dc99cf88_F00069	005	—	ab	ab
pdf-dc99cf88_F00070	005	—	$(\rho_Y)_{ab} = \sum_i \sqrt{\pi(x_i, y_a) \pi(x_i, y_b)}$	$(\rho_Y)_{ab} = \sum_i \sqrt{\pi(x_i, y_a) \pi(x_i, y_b)}$
pdf-dc99cf88_F00071	005	—	y_a	y_a
pdf-dc99cf88_F00072	005	—	y_b	y_b
pdf-dc99cf88_F00073	005	—	X	X
pdf-dc99cf88_F00074	005	—	$p, q: S \rightarrow \mathbb{R}$	$p, q: S \rightarrow \mathbb{R}$
pdf-dc99cf88_F00075	005	—	$B(p, q) := \sum_s \sqrt{p(s)q(s)}$	$B(p, q) := \sum_s \sqrt{p(s)q(s)}$
pdf-dc99cf88_F00076	005	—	$y_a \in Y$	$y_a \in Y$
pdf-dc99cf88_F00077	005	—	$\pi_a: X \rightarrow \mathbb{R}$	$\pi_a: X \rightarrow \mathbb{R}$
pdf-dc99cf88_F00078	005	—	$x_i \mapsto \pi(x_i \mid y_a)$	$x_i \mapsto \pi(x_i \mid y_a)$
pdf-dc99cf88_F00079	005	—	π_a	π_a
pdf-dc99cf88_F00080	005	—	π_b	π_b
pdf-dc99cf88_F00081	005	—	$(\rho_Y)_{ab} = \sqrt{\pi_Y(y_a) \pi_Y(y_b)} B(\pi_a, \pi_b)$	$(\rho_Y)_{ab} = \sqrt{\pi_Y(y_a) \pi_Y(y_b)} B(\pi_a, \pi_b)$
pdf-dc99cf88_F00082	005	—	$H(p, q) := \sqrt{1 - B(p, q)}$	$H(p, q) := \sqrt{1 - B(p, q)}$
pdf-dc99cf88_F00083	005	—	$\rho_X = \operatorname{tr}_Y \operatorname{Pr}_{\psi}$	$\rho_X = \operatorname{tr}_Y \operatorname{Pr}_{\psi}$
pdf-dc99cf88_F00084	005	—	$\pi_X: X \rightarrow \mathbb{R}$	$\pi_X: X \rightarrow \mathbb{R}$
pdf-dc99cf88_F00085	005	—	$(x, y) \in X \times Y$	$(x, y) \in X \times Y$
pdf-dc99cf88_F00086	005	—	x	x
pdf-dc99cf88_F00087	005	—	y	y
pdf-dc99cf88_F00088	005	—	ρ_X	ρ_X
pdf-dc99cf88_F00089	006	—	$\pi: X \times Y \rightarrow \mathbb{R}$	$\pi: X \times Y \rightarrow \mathbb{R}$
pdf-dc99cf88_F00090	006	—	y_c	y_c
pdf-dc99cf88_F00091	006	—	y_d	y_d
pdf-dc99cf88_F00092	006	—	$\pi(x_i, y_c) = \pi(x_i, y_d)$	$\pi(x_i, y_c) = \pi(x_i, y_d)$
pdf-dc99cf88_F00093	006	—	$\rho_Y = \operatorname{tr}_X Pr_{\psi}$	$\rho_Y = \operatorname{tr}_X Pr_{\psi}$
pdf-dc99cf88_F00094	006	—	$\pi(x_i, y_c) \neq \pi(x_i, y_d)$	$\pi(x_i, y_c) \neq \pi(x_i, y_d)$
pdf-dc99cf88_F00095	006	—	$\rho_Y(y_c) \neq \rho_Y(y_d)$	$\rho_Y(y_c) \neq \rho_Y(y_d)$
pdf-dc99cf88_F00096		—	$\square$	$\square$
pdf-dc99cf88_F00097	006	—	$\rho_Y(y_a)$	$\rho_Y(y_a)$
pdf-dc99cf88_F00098	006	—	$X = \{$	$X = \{$
pdf-dc99cf88_F00099	006	—	$\}$	$\}$
pdf-dc99cf88_F00100	006	—	$Y = \{$	$Y = \{$
pdf-dc99cf88_F00101	006	—	$T \subseteq X \times Y$	$T \subseteq X \times Y$
pdf-dc99cf88_F00102	006	—	T	T
pdf-dc99cf88_F00103	006	—	$\pi(x, y)$	$\pi(x, y)$
pdf-dc99cf88_F00104	006	—	$1 / 4$	$1/4$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
pdf-dc99cf88_F00105	006	—	$(x, y) \in T$	$(x, y) \in T$
pdf-dc99cf88_F00106	006	—	$\not= \rho_Y$	$\neq \rho_Y$
pdf-dc99cf88_F00107	006	—	$= \rho_X$	$= \rho_X$
pdf-dc99cf88_F00108	006	—	$\pi(x, y) = \frac{1}{ T }$	$\pi(x, y) = \frac{1}{ T }$
pdf-dc99cf88_F00109	006	—	$\pi(x, y) = 0$	$\pi(x, y) = 0$
pdf-dc99cf88_F00110	006	—	$\psi = \frac{1}{\sqrt{ T }} \sum_{(x, y) \in T} xy$	$\psi = \frac{1}{\sqrt{ T }} \sum_{(x, y) \in T} xy$
pdf-dc99cf88_F00111	006	—	$\rho_Y = \operatorname{tr}_X \operatorname{Pr}_\psi$	$\rho_Y = \operatorname{tr}_X \operatorname{Pr}_\psi$
pdf-dc99cf88_F00112	006	—	$\sum_i \psi_{ia} \psi_{ib} = d/ T $	$\sum_i \psi_{ia} \psi_{ib} = d/ T $
pdf-dc99cf88_F00113	006	—	d	d
pdf-dc99cf88_F00114	006	—	$x_i \in X$	$x_i \in X$
pdf-dc99cf88_F00115	006	—	$(x_i, y_a) \in T$	$(x_i, y_a) \in T$
pdf-dc99cf88_F00116	006	—	$(x_i, y_b) \in T$	$(x_i, y_b) \in T$
pdf-dc99cf88_F00117		—	Y	Y
pdf-dc99cf88_F00118		—	$\mathbb{C}^Y \cong \mathbb{C}^2$	$\mathbb{C}^Y \cong \mathbb{C}^2$
pdf-dc99cf88_F00119		—	$= \left[\begin{array}{ll} 1 & 0 \end{array} \right]^{\top}$	$= \left[\begin{array}{cc} 1 & 0 \end{array} \right]^{\top}$
pdf-dc99cf88_F00120		—	$= \left[\begin{array}{ll} 0 & 1 \end{array} \right]^{\top}$	$= \left[\begin{array}{cc} 0 & 1 \end{array} \right]^{\top}$
pdf-dc99cf88_F00121		—	$= \left[\begin{array}{llll} 1 & 0 & 0 & 0 \end{array} \right]^{\top}$	$= \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \end{array} \right]^{\top}$
pdf-dc99cf88_F00122	007	—	N	N
pdf-dc99cf88_F00123	007	—	$N \geq 2$	$N \geq 2$
pdf-dc99cf88_F00124	007	—	$S = X^{N-1} \times X$	$S = X^{N-1} \times X$
pdf-dc99cf88_F00125	007	—	$\left(x_{i_{N-1}}, \ldots, x_{i_2}, x_{i_1}\right)$	$\left(x_{i_{N-1}}, \ldots, x_{i_2}, x_{i_1}\right)$
pdf-dc99cf88_F00126	007	—	x_a	x_a
pdf-dc99cf88_F00127	007	—	$s = \left(x_{i_{N-1}} \cdots x_{i_2} x_{i_1}, x_a\right)$	$s = \left(x_{i_{N-1}} \cdots x_{i_2} x_{i_1}, x_a\right)$
pdf-dc99cf88_F00128	007	—	$x_{i_{N-1}} \cdots x_{i_2} x_{i_1}$	$x_{i_{N-1}} \cdots x_{i_2} x_{i_1}$
pdf-dc99cf88_F00129	007	—	i_1	i_1
pdf-dc99cf88_F00130	007	—	i_{N-1}	i_{N-1}
pdf-dc99cf88_F00131	007	—	$i, j, \ldots$	$i, j, \ldots$
pdf-dc99cf88_F00132	007	—	$a, b, \ldots$	$a, b, \ldots$
pdf-dc99cf88_F00133	007	—	X^k	X^k
pdf-dc99cf88_F00134	007	—	$k \geq 1$	$k \geq 1$
pdf-dc99cf88_F00135	007	—	$k = 1$	$k = 1$
pdf-dc99cf88_F00136	007	—	$\psi = \sum_{s \in S} \psi_s s$	$\psi = \sum_{s \in S} \psi_s s$
pdf-dc99cf88_F00137	007	—	$\psi_s = \sqrt{\pi(s)}$	$\psi_s = \sqrt{\pi(s)}$
pdf-dc99cf88_F00138	007	—	$\mathbb{C}^S \cong V^{\otimes N-1} \otimes V$	$\mathbb{C}^S \cong V^{\otimes N-1} \otimes V$
pdf-dc99cf88_F00139	007	—	$V = \mathbb{C}^X$	$V = \mathbb{C}^X$
pdf-dc99cf88_F00140	007	—	$s = x_{i_{N-1}} \cdots x_{i_1} x_a$	$s = x_{i_{N-1}} \cdots x_{i_1} x_a$
pdf-dc99cf88_F00141	007	—	$\psi_{i_{N-1} \cdots i_1 a} = \sqrt{\pi\left(x_{i_{N-1}}, \ldots, x_{i_1}, x_a\right)}$	$\psi_{i_{N-1} \cdots i_1 a} = \sqrt{\pi\left(x_{i_{N-1}}, \ldots, x_{i_1}, x_a\right)}$
pdf-dc99cf88_F00142	007	—	$V^{\otimes N-1} \otimes V$	$V^{\otimes N-1} \otimes V$
pdf-dc99cf88_F00143	007	—	$\rho_V = \operatorname{tr}_{V^{\otimes N-1}} \operatorname{Pr}_\psi: V \rightarrow V$	$\rho_V = \operatorname{tr}_{V^{\otimes N-1}} \operatorname{Pr}_\psi: V \rightarrow V$
pdf-dc99cf88_F00144	007	—	X^{N-1}	X^{N-1}
pdf-dc99cf88_F00145	007	—	$N = 5$	$N = 5$
pdf-dc99cf88_F00146	007	—	$x_{i_2} x_{i_1}$	$x_{i_2} x_{i_1}$
pdf-dc99cf88_F00147	007	—	i_2	i_2
pdf-dc99cf88_F00148	008	—	i_3	i_3
pdf-dc99cf88_F00149	008	—	i_4	i_4
pdf-dc99cf88_F00150	008	—	ρ_V	ρ_V
pdf-dc99cf88_F00151	008	—	$\hat{\rho}_{x_i x_{i_1}}$	$\hat{\rho}_{x_i x_{i_1}}$
pdf-dc99cf88_F00152	008	—	$\hat{\rho}_{x_{i_2} x_{i_1}}$	$\hat{\rho}_{x_{i_2} x_{i_1}}$
pdf-dc99cf88_F00153	008	—	$\rho_{x_{i_2} x_{i_1}}$	$\rho_{x_{i_2} x_{i_1}}$
pdf-dc99cf88_F00154	008	—	$\hat{\rho}_{x_{i_k} \cdots x_{i_1}}$	$\hat{\rho}_{x_{i_k} \cdots x_{i_1}}$
pdf-dc99cf88_F00155	008	—	$x_{i_k} \cdots x_{i_1}$	$x_{i_k} \cdots x_{i_1}$
pdf-dc99cf88_F00156	008	—	x_{i_1}	x_{i_1}
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
pdf-dc99cf88_F00157	008	—	$x_{i_1^{\prime}}$	$x_{i_1'}$
pdf-dc99cf88_F00158	008	—	$\hat{\rho}_{x_{i_1}}=\hat{\rho}_{x_{i_1}^{\prime}}$	$\hat{\rho}_{x_{i_1}} = \hat{\rho}_{x_{i_1}'}$
pdf-dc99cf88_F00159	008	—	$\psi_{i_{N-1}\cdots i_1}a=\psi_{i_{N-1}\cdots i_1^{\prime}}a$	$\psi_{i_{N-1}\cdots i_1}a = \psi_{i_{N-1}\cdots i_1'}a$
pdf-dc99cf88_F00160	008	—	$i_{N-1},\,\ldots,\,i_2$	$i_{N-1},\ldots,i_2$
pdf-dc99cf88_F00161	008	—	$1\leq k\leq N-1$	$1\leq k\leq N-1$
pdf-dc99cf88_F00162	008	—	$x_{i_1}=\text{d o g}$	$x_{i_1} = \textit{dog}$
pdf-dc99cf88_F00163	009	—	$x_{i_{N-1}}\cdots x_{i_1}$	$x_{i_{N-1}}\cdots x_{i_1}$
pdf-dc99cf88_F00164	009	—	$\hat{\rho}_{x_{i_{N-1}}\cdots x_{i_3}x_{i_2}x_{i_1}}$	$\hat{\rho}_{x_{i_{N-1}}\cdots x_{i_3}x_{i_2}x_{i_1}}$
pdf-dc99cf88_F00165	009	—	$A\times B\times C\times D$	$A\times B\times C\times D$
pdf-dc99cf88_F00166	009	—	$X\times X\times X\times X$	$X\times X\times X\times X$
pdf-dc99cf88_F00167	009	—	2×2	2×2
pdf-dc99cf88_F00168	009	—	$A=\{$	$A = \{$
pdf-dc99cf88_F00169	009	—	$B=\{$	$B = \{$
pdf-dc99cf88_F00170	009	—	$C=\{\mathrm{dog}$	$C = \{\text{dog}$
pdf-dc99cf88_F00171	009	—	$D=$	$D =$
pdf-dc99cf88_F00172	009	—	$N=4$	$N = 4$
pdf-dc99cf88_F00173	009	—	$\pi:A\times B\times C\times D\rightarrow\mathbb{R}$	$\pi:A\times B\times C\times D\rightarrow\mathbb{R}$
pdf-dc99cf88_F00174	009	—	$T\subseteq A\times B\times C\times D$	$T\subseteq A\times B\times C\times D$
pdf-dc99cf88_F00175	009	—	$\mathbb{C}^A\otimes\mathbb{C}^B\otimes\mathbb{C}^C\otimes\mathbb{C}^D$	$\mathbb{C}^A\otimes\mathbb{C}^B\otimes\mathbb{C}^C\otimes\mathbb{C}^D$
pdf-dc99cf88_F00176	009	—	$\mathbb{C}^2$	$\mathbb{C}^2$
pdf-dc99cf88_F00177	009	—	$A\times B\times C$	$A\times B\times C$
pdf-dc99cf88_F00178	009	—	D	D
pdf-dc99cf88_F00179	009	—	$\rho_D=\operatorname{tr}_{A\times B\times C}\operatorname{Pr}_{\psi}$	$\rho_D = \operatorname{tr}_{A\times B\times C}\operatorname{Pr}_{\psi}$
pdf-dc99cf88_F00180	017	—	$\hat{\rho}_{x_{i_1}}$	$\hat{\rho}_{x_{i_1}}$
pdf-dc99cf88_F00181		—	$V^{\otimes N-1}\rightarrow V$	$V^{\otimes N-1}\rightarrow V$
pdf-dc99cf88_F00182	014	—	$\psi\in V^{\otimes N-1}\otimes V$	$\psi\in V^{\otimes N-1}\otimes V$
pdf-dc99cf88_F00183	010	—	$\left[\begin{array}{l}1\\0\end{array}\right]$	$\left[\begin{array}{c}1\\0\end{array}\right]$
pdf-dc99cf88_F00184	010	—	$\left[\begin{array}{l}0\\1\end{array}\right]$	$\left[\begin{array}{c}0\\1\end{array}\right]$
pdf-dc99cf88_F00185	010	—	$1\,/\,5$	$1/5$
pdf-dc99cf88_F00186	010	—	ρ_D	ρ_D
pdf-dc99cf88_F00187	010	—	$(a,\,b,\,c)\,\,\text{in}\,\,T$	$(a,b,c)\in T$
pdf-dc99cf88_F00188	010	—	$\hat{\rho}_{\text{\text{dog}}}$	$\hat{\rho}_{\text{dog}}$
pdf-dc99cf88_F00189	010	—	dog	dog
pdf-dc99cf88_F00190	010	—	$c=\text{d o g}$	$c = \textit{dog}$
pdf-dc99cf88_F00191	010	—	$(a,\,b)\,\,\text{in}\,\,A\times B$	$(a,b)\in A\times B$
pdf-dc99cf88_F00192	010	—	$(a,\,b)$	(a,b)
pdf-dc99cf88_F00193	010	—	$b=$	$b =$
pdf-dc99cf88_F00194	010	—	$a\,\,\text{in}\,\,A$	$a\in A$
pdf-dc99cf88_F00195	010	—	$a=$	$a =$
pdf-dc99cf88_F00196	010	—	d o g	$\textit{dog}$
pdf-dc99cf88_F00197	010	—	$\hat{\rho}_{\text{\text{dog}}}=\hat{\rho}_{\text{\text{black dog}}}+\hat{\rho}_{\text{\text{white dog}}}$	$\hat{\rho}_{\text{dog}} = \hat{\rho}_{\text{black dog}} + \hat{\rho}_{\text{white dog}}$
pdf-dc99cf88_F00198	010	—	$\hat{\rho}_{\text{\text{black dog}}} =$	$\hat{\rho}_{\text{black dog}} =$
pdf-dc99cf88_F00199	010	—	$\hat{\rho}_{\text{\text{small black dog}}}+\hat{\rho}_{\text{\text{big black dog}}}$	$\hat{\rho}_{\text{small black dog}} + \hat{\rho}_{\text{big black dog}}$
pdf-dc99cf88_F00200	010	—	$\hat{\rho}_{\text{\text{white dog}}}=\hat{\rho}_{\text{\text{small white dog}}}$	$\hat{\rho}_{\text{white dog}} = \hat{\rho}_{\text{small white dog}}$
pdf-dc99cf88_F00201	011	—	$\pi(x_{i_k},\,\cdots,\,x_{i_1})$	$\pi(x_{i_k},\ldots,x_{i_1})$
pdf-dc99cf88_F00202	011	—	$T\subseteq X^{N-1}\times X$	$T\subseteq X^{N-1}\times X$
pdf-dc99cf88_F00203	011	—	$ T $	$ T $
pdf-dc99cf88_F00204	011	—	$\pi(x_{i_k}\cdots x_{i_1}x_a\mid x_{i_k}\cdots x_{i_1})$	$\pi(x_{i_k}\cdots x_{i_1}x_a\mid x_{i_k}\cdots x_{i_1})$
pdf-dc99cf88_F00205	011	—	$x_a\in X$	$x_a\in X$
pdf-dc99cf88_F00206	011	—	$\rho_{x_{i_k}\cdots x_{i_1}}$	$\rho_{x_{i_k}\cdots x_{i_1}}$
pdf-dc99cf88_F00207	011	—	$\left\langle\rho_{x_{i_k}\cdots x_{i_1}}x_a,x_a\right\rangle$	$\left\langle\rho_{x_{i_k}\cdots x_{i_1}}x_a,x_a\right\rangle$
pdf-dc99cf88_F00208	011	—	$\operatorname{tr}\hat{\rho}_{\text{\text{dog}}}=3\,/\,5=\pi(\operatorname{dog})=\sum_{a,b,d}\pi(a,b,\text{\text{dog}},d)$	$\operatorname{tr}\hat{\rho}_{\text{dog}} = 3/5 = \pi(\text{dog}) = \sum_{a,b,d}\pi(a,b,\text{dog},d)$
pdf-dc99cf88_F00209	011	—	$\rho_{\text{\text{dog}}}$	ρ_{dog}
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
pdf-dc99cf88_F00210	011	—	$\left\langle \text{\texttt{\textbackslash langle\rho_{\text {dog }}}} \right\rangle.$	$\langle \rho_{\text{dog}}$
pdf-dc99cf88_F00211	011	—	$\text{\texttt{\textbackslash rangle}}=2 / 3=\pi$	$\rangle = 2/3 = \pi$
pdf-dc99cf88_F00212	011	—	$\text{\texttt{\textbackslash mid}}$	$ $
pdf-dc99cf88_F00213	011	—	$(a, b, c, d) \text{\texttt{\textbackslash in}} T$	$(a, b, c, d) \in T$
pdf-dc99cf88_F00214	011	—	$d=\text{b a r k s}$	$d = barks$
pdf-dc99cf88_F00215	011	—	$\text{\texttt{\textbackslash rangle}}=1 / 3=\pi$	$\rangle = 1/3 = \pi$
pdf-dc99cf88_F00216	011	—	$\text{\texttt{\textbackslash mid}} \text{\texttt{\textbackslash operatorname}}\{\text{dog}\}$	$ \text{ dog}$
pdf-dc99cf88_F00217	011	—	$\text{\texttt{\textbackslash rho_{\mathrm{dog}}}}$	ρ_{dog}
pdf-dc99cf88_F00218	011	—	$\text{\texttt{\textbackslash rho_{\text {black dog }}}}$	$\rho_{\text{black dog}}$
pdf-dc99cf88_F00219	011	—	$\text{\texttt{\textbackslash rho_{\text {white dog }}}}$	$\rho_{\text{white dog}}$
pdf-dc99cf88_F00220	011	—	$k+1$	$k + 1$
pdf-dc99cf88_F00221	011	—	$\text{\texttt{\textbackslash hat{\rho}_{x_{\text {\scriptsize i}}\{k\}} \cdots x_{\text {\scriptsize i}}\{1\}}}}=\text{\texttt{\textbackslash sum_{i_{k+1}}}} \text{\texttt{\textbackslash hat{\rho}_{x_{\text {\scriptsize i}}\{k+1\}} x_{\text {\scriptsize i}}\{k\}} \cdots x_{\text {\scriptsize i}}\{1\}}}}$	$\hat{\rho}_{x_{i_k} \cdots x_{i_1}} = \sum_{i_{k+1}} \hat{\rho}_{x_{i_{k+1}} x_{i_k} \cdots x_{i_1}}$
pdf-dc99cf88_F00222	012	—	$N-1$	$N - 1$
pdf-dc99cf88_F00223	012	—	$\text{\texttt{\textbackslash rho_{x_{\text {\scriptsize i}}\{N-1\}} \cdots x_{\text {\scriptsize i}}\{3\}} x_{\text {\scriptsize i}}\{2\}} x_{\text {\scriptsize i}}\{1\}}}$	$\rho_{x_{i_{N-1}} \cdots x_{i_3} x_{i_2} x_{i_1}}$
pdf-dc99cf88_F00224	012	—	$\text{\texttt{\textbackslash rho_{x_{\text {\scriptsize i}}\{1\}}}}$	$\rho_{x_{i_1}}$
pdf-dc99cf88_F00225	012	—	$\text{\texttt{\textbackslash pi}}($	$\pi($
pdf-dc99cf88_F00226	012	—	$\text{\texttt{\textbackslash mid}} \text{\texttt{\textbackslash operatorname}}\{\text{dog}\})=1 / 3$	$ \text{ dog}) = 1/3$
pdf-dc99cf88_F00227	012	—	$\text{\texttt{\textbackslash mid}} \text{\texttt{\textbackslash operatorname}}\{\text{dog}\})=2 / 3$	$ \text{ dog}) = 2/3$
pdf-dc99cf88_F00228	012	—	$\text{\texttt{\textbackslash hat{\rho}_{\text {\scriptsize i}}\{\text {dog }}}}=\text{\texttt{\textbackslash hat{\rho}_{\text {\scriptsize i}}\{\text {white dog }}}}+\text{\texttt{\textbackslash hat{\rho}_{\text {\scriptsize i}}\{\text {black dog }}}}$	$\hat{\rho}_{\text{dog}} = \hat{\rho}_{\text{white dog}} + \hat{\rho}_{\text{black dog}}$
pdf-dc99cf88_F00229	013	—	$\text{\texttt{\textbackslash hat{\rho}_{\text {\scriptsize i}}\{\text {dog }}}}=\text{\texttt{\textbackslash hat{\rho}_{\text {\scriptsize i}}\{\text {small black dog }}}}+\text{\texttt{\textbackslash hat{\rho}_{\text {\scriptsize i}}\{\text {big black dog }}}}+\text{\texttt{\textbackslash hat{\rho}_{\text {\scriptsize i}}\{\text {small white dog }}}}$	$\hat{\rho}_{\text{dog}} = \hat{\rho}_{\text{small black dog}} + \hat{\rho}_{\text{big black dog}} + \hat{\rho}_{\text{small white dog}}$
pdf-dc99cf88_F00230	013	—	$V \text{\texttt{\textbackslash otimes}} V$	$V \otimes V$
pdf-dc99cf88_F00231	013	—	$x_{\text {\scriptsize i}}\{3\} x_{\text {\scriptsize i}}\{2\} x_{\text {\scriptsize i}}\{1\}$	$x_{i_3} x_{i_2} x_{i_1}$
pdf-dc99cf88_F00232	013	—	$x_{\text {\scriptsize j}}\{2\} x_{\text {\scriptsize j}}\{1\}$	$x_{j_2} x_{j_1}$
pdf-dc99cf88_F00233	013	—	$\text{\texttt{\textbackslash hat{\rho}_{x_{\text {\scriptsize i}}\{3\}} x_{\text {\scriptsize i}}\{2\}} x_{\text {\scriptsize i}}\{1\}}}$	$\hat{\rho}_{x_{i_3} x_{i_2} x_{i_1}}$
pdf-dc99cf88_F00234	013	—	$\text{\texttt{\textbackslash hat{\rho}_{x_{\text {\scriptsize j}}\{2\}} x_{\text {\scriptsize j}}\{1\}}}$	$\hat{\rho}_{x_{j_2} x_{j_1}}$
pdf-dc99cf88_F00235	013	—	$\text{\texttt{\textbackslash hat{\rho}_{\text {\scriptsize i}}\{\text {tiny toy poodle }}}}$	$\hat{\rho}_{\text{tiny toy poodle}}$
pdf-dc99cf88_F00236	013	—	$\text{\texttt{\textbackslash hat{\rho}_{\text {\scriptsize i}}\{\text {new park }}}}$	$\hat{\rho}_{\text{new park}}$
pdf-dc99cf88_F00237	013	—	$\text{\texttt{\textbackslash hat{\rho}_{\mathrm{I}} \text {\texttt{\textbackslash text { walked my tiny toy poodle at the new park }}}}}$	$\hat{\rho}_{\text{I walked my tiny toy poodle at the new park}}$
pdf-dc99cf88_F00238	013	—	$\text{\texttt{\textbackslash rho^{\prime}}}$	ρ'
pdf-dc99cf88_F00239	013	—	$\text{\texttt{\textbackslash rho}} \text{\texttt{\textbackslash geq}} \text{\texttt{\textbackslash rho^{\prime}}}$	$\rho \geq \rho'$
pdf-dc99cf88_F00240	013	—	$\text{\texttt{\textbackslash rho}}-\text{\texttt{\textbackslash rho^{\prime}}}$	$\rho - \rho'$
pdf-dc99cf88_F00241	013	—	$x_{\text {\scriptsize i}}\{k+1\}$	$x_{i_{k+1}}$
pdf-dc99cf88_F00242	013	—	$\text{\texttt{\textbackslash hat{\rho}_{\text {\scriptsize i}}\{\text {dog }}}} \text{\texttt{\textbackslash geq}} \text{\texttt{\textbackslash hat{\rho}_{\text {\scriptsize i}}\{\text {black dog }}}}$	$\hat{\rho}_{\text{dog}} \geq \hat{\rho}_{\text{black dog}}$
pdf-dc99cf88_F00243	013	—	$\text{\texttt{\textbackslash hat{\rho}_{\text {\scriptsize i}}\{\text {dog }}}} \text{\texttt{\textbackslash geq}} \text{\texttt{\textbackslash hat{\rho}_{\text {\scriptsize i}}\{\text {white dog }}}}$	$\hat{\rho}_{\text{dog}} \geq \hat{\rho}_{\text{white dog}}$
pdf-dc99cf88_F00244	014	—	$s^{\text{\texttt{\textbackslash prime}}}$	s'
pdf-dc99cf88_F00245	014	—	$\text{\texttt{\textbackslash rho_{s}}} \text{\texttt{\textbackslash geq}} \pi(\text{\texttt{\textbackslash prime}} \text{\texttt{\textbackslash mid}} s \text{\texttt{\textbackslash right}}) \text{\texttt{\textbackslash rho_{s^{\text {\scriptsize i}}\{\text {\texttt{\textbackslash prime}}}}}}}$	$\rho_s \geq \pi(s' \mid s) \rho_{s'}$
pdf-dc99cf88_F00246	014	—	$\text{\texttt{\textbackslash hat{\rho}_{\text {\scriptsize i}}\{s\}}} \text{\texttt{\textbackslash geq}} \text{\texttt{\textbackslash hat{\rho}_{\text {\scriptsize i}}\{s^{\text {\scriptsize i}}\{\text {\texttt{\textbackslash prime}}}}}}}$	$\hat{\rho}_s \geq \hat{\rho}_{s'}$
pdf-dc99cf88_F00247	014	—	$N \text{\texttt{\textbackslash geq}} 1$	$N \geq 1$
pdf-dc99cf88_F00248	014	—	L	L
pdf-dc99cf88_F00249	014	—	$\text{\texttt{\textbackslash sqcup_{k=1}^{\text {\scriptsize N}-1}} X^k \text {\texttt{\textbackslash times}} X}$	$\sqcup_{k=1}^{N-1} X^k \times X$
pdf-dc99cf88_F00250	014	—	$s=\text{\texttt{\textbackslash left}}(x_{\text {\scriptsize i}}\{k\}, \text{\texttt{\textbackslash ldots}}, x_{\text {\scriptsize i}}\{1\}, x_{\text {\scriptsize a}}\text{\texttt{\textbackslash right}})$	$s = (x_{i_k}, \dots, x_{i_1}, x_a)$
pdf-dc99cf88_F00251	014	—	$k \text{\texttt{\textbackslash leq}} N-1$	$k \leq N - 1$
pdf-dc99cf88_F00252	014	—	$x_{\text {\scriptsize i}}\{k\} \cdots x_{\text {\scriptsize i}}\{1\} x_{\text {\scriptsize a}}$	$x_{i_k} \cdots x_{i_1} x_a$
pdf-dc99cf88_F00253	014	—	$\text{\texttt{\textbackslash left}}(x_{\text {\scriptsize i}}\{k\}, \text{\texttt{\textbackslash ldots}}, x_{\text {\scriptsize i}}\{1\}, x_{\text {\scriptsize a}}\text{\texttt{\textbackslash right}})$	$(x_{i_k}, \dots, x_{i_1}, x_a)$
pdf-dc99cf88_F00254	014	—	$s, s^{\text{\texttt{\textbackslash prime}}} \text{\texttt{\textbackslash in}} L$	$s, s' \in L$
pdf-dc99cf88_F00255	014	—	$s \text{\texttt{\textbackslash leq}} s^{\text{\texttt{\textbackslash prime}}}$	$s \leq s'$
pdf-dc99cf88_F00256	014	—	$d \text{ o g } \text{\texttt{\textbackslash leq}}$	$dog \leq$
pdf-dc99cf88_F00257	014	—	$s \text{\texttt{\textbackslash in}} L$	$s \in L$
pdf-dc99cf88_F00258	014	—	$s \text{\texttt{\textbackslash leq}} s$	$s \leq s$
pdf-dc99cf88_F00259	014	—	$s, s^{\text{\texttt{\textbackslash prime}}}, s^{\text{\texttt{\textbackslash prime \text {\scriptsize i}}\{\text {\texttt{\textbackslash prime}}}}}$ \text{\texttt{\textbackslash in}} L	$s, s', s'' \in L$
pdf-dc99cf88_F00260	014	—	$s^{\text{\texttt{\textbackslash prime}}} \text{\texttt{\textbackslash leq}} s^{\text{\texttt{\textbackslash prime \text {\scriptsize i}}\{\text {\texttt{\textbackslash prime}}}}}$	$s' \leq s''$
pdf-dc99cf88_F00261	014	—	$s^{\text{\texttt{\textbackslash prime \text {\scriptsize i}}\{\text {\texttt{\textbackslash prime}}}}}$	s''
pdf-dc99cf88_F00262	014	—	$s \text{\texttt{\textbackslash leq}} s^{\text{\texttt{\textbackslash prime \text {\scriptsize i}}\{\text {\texttt{\textbackslash prime}}}}}$	$s \leq s''$
pdf-dc99cf88_F00263	014	—	$X^{\text{\texttt{\textbackslash N}}} \text{\texttt{\textbackslash cong}} X^{\text{\texttt{\textbackslash N}-1}} \text{\texttt{\textbackslash times}} X$	$X^N \cong X^{N-1} \times X$
pdf-dc99cf88_F00264	014	—	$\text{\texttt{\textbackslash pi}}\text{\texttt{\textbackslash colon}} X^{\text{\texttt{\textbackslash N}-1}} \text{\texttt{\textbackslash times}} X \text{\texttt{\textbackslash rrightarrow}} \text{\texttt{\textbackslash mathbb{R}}}$	$\pi : X^{N-1} \times X \rightarrow \mathbb{R}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
pdf-dc99cf88_F00265	014	—	$s \in X^{N-1}$	$s \in X^{N-1}$
pdf-dc99cf88_F00266	014	—	$s=x_{\{i_k\}} \cdots x_{\{i_1\}}$	$s = x_{i_k} \cdots x_{i_1}$
pdf-dc99cf88_F00267	014	—	$\mathcal{L} \subseteq L$	$\mathcal{L} \subseteq L$
pdf-dc99cf88_F00268	014	—	$(\operatorname{Pos}(V), \leq)$	$(\operatorname{Pos}(V), \leq)$
pdf-dc99cf88_F00269	014	—	$(\mathcal{L}, \leq) \rightarrow (\operatorname{Pos}(V), \leq)$	$(\mathcal{L}, \leq) \rightarrow (\operatorname{Pos}(V), \leq)$
pdf-dc99cf88_F00270	015	—	$s^{\{\prime\}}=x_{\{i_m\}} \cdots x_{\{i_{k+1}\}} s$	$s' = x_{i_m} \cdots x_{i_{k+1}} s$
pdf-dc99cf88_F00271	015	—	$1 \leq k \leq m \leq N-1$	$1 \leq k \leq m \leq N-1$
pdf-dc99cf88_F00272	015	—	$\hat{\rho}_{\text{black cat}} \leq \hat{\rho}_{\text{dog}}$	$\hat{\rho}_{\text{black cat}} \leq \hat{\rho}_{\text{dog}}$
pdf-dc99cf88_F00273	015	—	$\hat{\rho}_{\text{black cat}}=\frac{1}{5}\left[\begin{array}{ll}\theta & 0 \\ 0 & 1\end{array}\right]$	$\hat{\rho}_{\text{black cat}} = \frac{1}{5} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$
pdf-dc99cf88_F00274	015	—	$\operatorname{dog} \not\leq$	$\operatorname{dog} \not\leq$
pdf-dc99cf88_F00275	015	—	$\mathcal{L}$	$\mathcal{L}$
pdf-dc99cf88_F00276	015	—	$\operatorname{by}^3 \uparrow \operatorname{dog} \coloneqq \{s \in L \mid \operatorname{dog} \leq s\}$	$\operatorname{by}^3 \uparrow \{\operatorname{dog}\} := \{s \in L \mid \operatorname{dog} \leq s\}$
pdf-dc99cf88_F00277	015	—	$x_{\{i_k\}} \cdots x_{\{i_2\}} x_{\{i_1\}} \in \mathcal{L} \subseteq L$	$x_{i_k} \cdots x_{i_2} x_{i_1} \in \mathcal{L} \subseteq L$
pdf-dc99cf88_F00278	015	—	$x_{\{i_1\}}=\operatorname{dog}$	$x_{i_1} = \operatorname{dog}$
pdf-dc99cf88_F00279	015	—	$\uparrow \operatorname{dog}$	$\uparrow \{\operatorname{dog}\}$
pdf-dc99cf88_F00280	015	—	$\mathcal{L} \rightarrow \operatorname{Pos}(V)$	$\mathcal{L} \rightarrow \operatorname{Pos}(V)$
pdf-dc99cf88_F00281	015	—	$(P, \leq, \cdot, 1)$	$(P, \leq, \cdot, 1)$
pdf-dc99cf88_F00282	015	—	$(P, \leq)$	$(P, \leq)$
pdf-dc99cf88_F00283		—	$1 \in P$	$1 \in P$
pdf-dc99cf88_F00284		—	$L^{\operatorname{op}} \rightarrow UL$	$L^{\operatorname{op}} \rightarrow UL$
pdf-dc99cf88_F00285		—	UL	UL
pdf-dc99cf88_F00286		—	$\cdot : P \times P \rightarrow P$	$\cdot : P \times P \rightarrow P$
pdf-dc99cf88_F00287	016	—	$p \, q$	pq
pdf-dc99cf88_F00288	016	—	$p \cdot q$	$p \cdot q$
pdf-dc99cf88_F00289		—	$p, p^{\{\prime\}}, q, q^{\{\prime\}} \in P$	$p, p', q, q' \in P$
pdf-dc99cf88_F00290		—	$p \leq p^{\{\prime\}}$	$p \leq p'$
pdf-dc99cf88_F00291		—	$q \leq q^{\{\prime\}}$	$q \leq q'$
pdf-dc99cf88_F00292		—	$p \, q \leq p^{\{\prime\}} q^{\{\prime\}}$	$pq \leq p' q'$
pdf-dc99cf88_F00293		—	$1 \, p=p \, 1=p$	$1p = p1 = p$
pdf-dc99cf88_F00294		—	$p \in P$	$p \in P$
pdf-dc99cf88_F00295		—	$(p \, q) \, r=p(q \, r)$	$(pq)r = p(qr)$
pdf-dc99cf88_F00296		—	$p, q, r \in P$	$p, q, r \in P$
pdf-dc99cf88_F00297		—	$p \, q=q \, p$	$pq = qp$
pdf-dc99cf88_F00298		—	$p, q \in P$	$p, q \in P$
pdf-dc99cf88_F00299	016	—	$[0,1] \subseteq \mathbb{R}$	$[0,1] \subseteq \mathbb{R}$
pdf-dc99cf88_F00300	016	—	$(\mathcal{V}, \leq, \cdot, 1)$	$(\mathcal{V}, \leq, \cdot, 1)$
pdf-dc99cf88_F00301	016	—	$\mathcal{V}$	$\mathcal{V}$
pdf-dc99cf88_F00302	016	—	$\mathcal{C}$	$\mathcal{C}$
pdf-dc99cf88_F00303		—	$\operatorname{ob}(\mathcal{C})$	$\operatorname{ob}(\mathcal{C})$
pdf-dc99cf88_F00304		—	$c, d, \ldots$	$c, d, \dots$
pdf-dc99cf88_F00305		—	$\mathcal{C}(c, d) \in \mathcal{V}$	$\mathcal{C}(c, d) \in \mathcal{V}$
pdf-dc99cf88_F00306		—	c	c
pdf-dc99cf88_F00307		—	$1 \leq \mathcal{C}(c, c)$	$1 \leq \mathcal{C}(c, c)$
pdf-dc99cf88_F00308		—	$\mathcal{C}(c, d) \cdot \mathcal{C}(d, e) \leq \mathcal{C}(c, e)$	$\mathcal{C}(c, d) \cdot \mathcal{C}(d, e) \leq \mathcal{C}(c, e)$
pdf-dc99cf88_F00309		—	c, d, e	c, d, e
pdf-dc99cf88_F00310	016	—	$\mathcal{D}$	$\mathcal{D}$
pdf-dc99cf88_F00311	016	—	$f: \operatorname{ob}(\mathcal{C}) \rightarrow \operatorname{ob}(\mathcal{D})$	$f: \operatorname{ob}(\mathcal{C}) \rightarrow \operatorname{ob}(\mathcal{D})$
pdf-dc99cf88_F00312	016	—	$c, c^{\{\prime\}}$	c, c'
pdf-dc99cf88_F00313	016	—	$\mathcal{L} \subseteq L \subseteq \sqcup_{k=1}^{N-1} X^k \times X$	$\mathcal{L} \subseteq L \subseteq \sqcup_{k=1}^{N-1} X^k \times X$
pdf-dc99cf88_F00314	016	—	$s, s^{\{\prime\}} \in \mathcal{L}$	$s, s' \in \mathcal{L}$
pdf-dc99cf88_F00315	016	—	$1 \leq \mathcal{L}(s, s)$	$1 \leq \mathcal{L}(s, s)$
pdf-dc99cf88_F00316	016	—	$\mathcal{L}(s, s^{\{\prime\}}) \mathcal{L}(s^{\{\prime\}}, s^{\{\prime\prime\}}) \leq \mathcal{L}(s, s^{\{\prime\prime\}})$	$\mathcal{L}(s, s') \mathcal{L}(s', s'') \leq \mathcal{L}(s, s'')$
pdf-dc99cf88_F00317	016	—	$s, s^{\{\prime\}}$	s, s'
pdf-dc99cf88_F00318	016	—	$s \in \mathcal{L}$	$s \in \mathcal{L}$
pdf-dc99cf88_F00319	016	—	$\mathcal{D} \subseteq \operatorname{Pos}(V)$	$\mathcal{D} \subseteq \operatorname{Pos}(V)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value				

Identifier	Page	Conf.	LaTeX source	Rendered
pdf-dc99cf88_F00320	016	—	$\hat{\rho}_s, \hat{\rho}_{s'} \in \mathcal{D}$	$\hat{\rho}_s, \hat{\rho}_{s'} \in \mathcal{D}$
pdf-dc99cf88_F00321		—	$\pi \left(s^{\prime} \mid s \right)$	$\pi \left(s^{\prime} \mid s \right)$
pdf-dc99cf88_F00322		—	$\pi (t \, s \, \mid s)$	$\pi (ts \mid s)$
pdf-dc99cf88_F00323		—	$s^{\prime} = t \, s$	$s^{\prime} = ts$
pdf-dc99cf88_F00324		—	t	t
pdf-dc99cf88_F00325		—	$\operatorname{dog} \mid \operatorname{dog}$	$\operatorname{dog} \mid \operatorname{dog}$
pdf-dc99cf88_F00326		—	$\operatorname{dog}) = \pi(\operatorname{black} \operatorname{dog}) / \pi(\operatorname{dog})$	$\operatorname{dog}) = \pi(\operatorname{black} \operatorname{dog}) / \pi(\operatorname{dog})$
pdf-dc99cf88_F00327	017	—	$1 \leq \mathcal{D} \left(\hat{\rho}_s, \hat{\rho}_s \right)$	$1 \leq \mathcal{D} \left(\hat{\rho}_s, \hat{\rho}_s \right)$
pdf-dc99cf88_F00328	017	—	$\mathcal{D} \left(\hat{\rho}_s, \hat{\rho}_{s'} \right) \mathcal{D} \left(\hat{\rho}_{s'}, \hat{\rho}_{s''} \right) \leq$	$\mathcal{D} \left(\hat{\rho}_s, \hat{\rho}_{s'} \right) \mathcal{D} \left(\hat{\rho}_{s'}, \hat{\rho}_{s''} \right) \leq$
pdf-dc99cf88_F00329	017	—	$\mathcal{D} \left(\hat{\rho}_s, \hat{\rho}_{s''} \right)$	$\mathcal{D} \left(\hat{\rho}_s, \hat{\rho}_{s''} \right)$
pdf-dc99cf88_F00330	017	—	$\operatorname{tr} \hat{\rho}_s = \pi(s)$	$\operatorname{tr} \hat{\rho}_s = \pi(s)$
pdf-dc99cf88_F00331	017	—	$\operatorname{tr} \hat{\rho}_{s'} / \operatorname{tr} \hat{\rho}_s = \pi \left(s^{\prime} \right) / \pi(s) = \pi \left(s^{\prime} \mid s \right)$	$\operatorname{tr} \hat{\rho}_{s'} / \operatorname{tr} \hat{\rho}_s = \pi \left(s^{\prime} \right) / \pi(s) = \pi \left(s^{\prime} \mid s \right)$
pdf-dc99cf88_F00332	017	—	$\mathcal{L} \left(s, s^{\prime} \right) \leq \mathcal{D} \left(\hat{\rho}_s, \hat{\rho}_{s'} \right)$	$\mathcal{L} \left(s, s^{\prime} \right) \leq \mathcal{D} \left(\hat{\rho}_s, \hat{\rho}_{s'} \right)$
pdf-dc99cf88_F00333	017	—	$\mathcal{L} \rightarrow \mathcal{D}$	$\mathcal{L} \rightarrow \mathcal{D}$
pdf-dc99cf88_F00334	017	—	$\mathcal{L} \left(s, s^{\prime} \right) = \pi \left(s^{\prime} \mid s \right)$	$\mathcal{L} \left(s, s^{\prime} \right) = \pi \left(s^{\prime} \mid s \right)$
pdf-dc99cf88_F00335	017	—	$\hat{\rho}_{s'} \leq \hat{\rho}_s$	$\hat{\rho}_{s'} \leq \hat{\rho}_s$
pdf-dc99cf88_F00336	017	—	$\mathcal{D}^{\operatorname{op}} \left(\hat{\rho}_{s'}, \hat{\rho}_s \right) :=$	$\mathcal{D}^{\operatorname{op}} \left(\hat{\rho}_{s'}, \hat{\rho}_s \right) :=$
pdf-dc99cf88_F00337	017	—	$\mathcal{D} \left(\hat{\rho}_s, \hat{\rho}_{s'} \right) = \operatorname{tr} \hat{\rho}_{s'} / \operatorname{tr} \hat{\rho}_s = \pi \left(s^{\prime} \mid s \right)$	$\mathcal{D} \left(\hat{\rho}_s, \hat{\rho}_{s'} \right) = \operatorname{tr} \hat{\rho}_{s'} / \operatorname{tr} \hat{\rho}_s = \pi \left(s^{\prime} \mid s \right)$
pdf-dc99cf88_F00338	017	—	$\langle \hat{\rho}_{s'} x, x \rangle = \pi \left(s^{\prime} x \right) \leq \pi(sx) =$	$\langle \hat{\rho}_{s'} x, x \rangle = \pi \left(s^{\prime} x \right) \leq \pi(sx) =$
pdf-dc99cf88_F00339	017	—	$\langle \hat{\rho}_s x, x \rangle$	$\langle \hat{\rho}_s x, x \rangle$
pdf-dc99cf88_F00340	017	—	$\pi \left(s^{\prime} x \right) > 0$	$\pi \left(s^{\prime} x \right) > 0$
pdf-dc99cf88_F00341	017	—	$\pi(sx) > 0$	$\pi(sx) > 0$
pdf-dc99cf88_F00342	017	—	$x \mapsto -\log(x)$	$x \mapsto -\log(x)$
pdf-dc99cf88_F00343	017	—	$[0,1] \rightarrow [0,\infty]$	$[0,1] \rightarrow [0,\infty]$
pdf-dc99cf88_F00344	018	—	$\psi_{\operatorname{net}}$	$\psi_{\operatorname{net}}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value				

Unrecovered image regions — TikZ / table / failed-math candidates

Regions MathPix left as images (no LaTeX). Reconstruct with pdfdrill vision (LLM classify + transcribe to TikZ/tabular/math); verify an LLM result against the real ink with inkdrill.

Identifier	Page	Image	Source
pdf-dc99cf88_DIA_0001	008		tex.zip (local)
pdf-dc99cf88_DIA_0002	009		tex.zip (local)
pdf-dc99cf88_DIA_0003	013		tex.zip (local)
pdf-dc99cf88_DIA_0004	017		tex.zip (local)